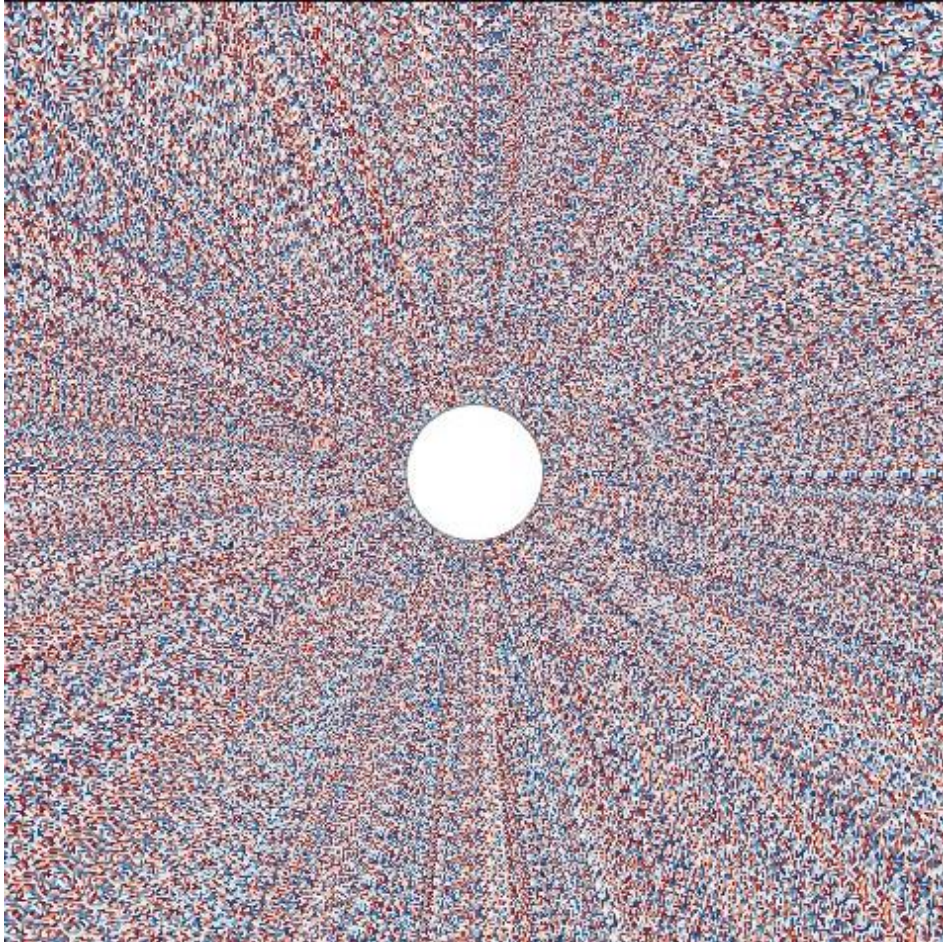


The reflex instability

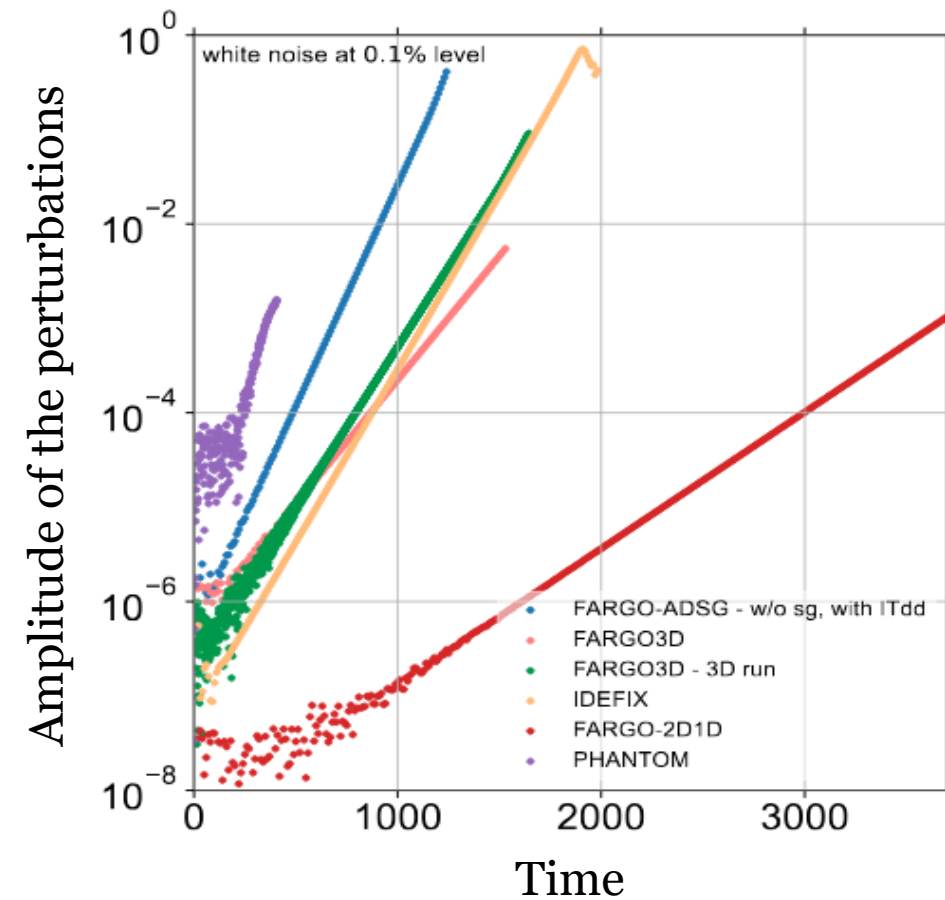
or how not to model the indirect term



The reflex motion of the central star induces a global instability

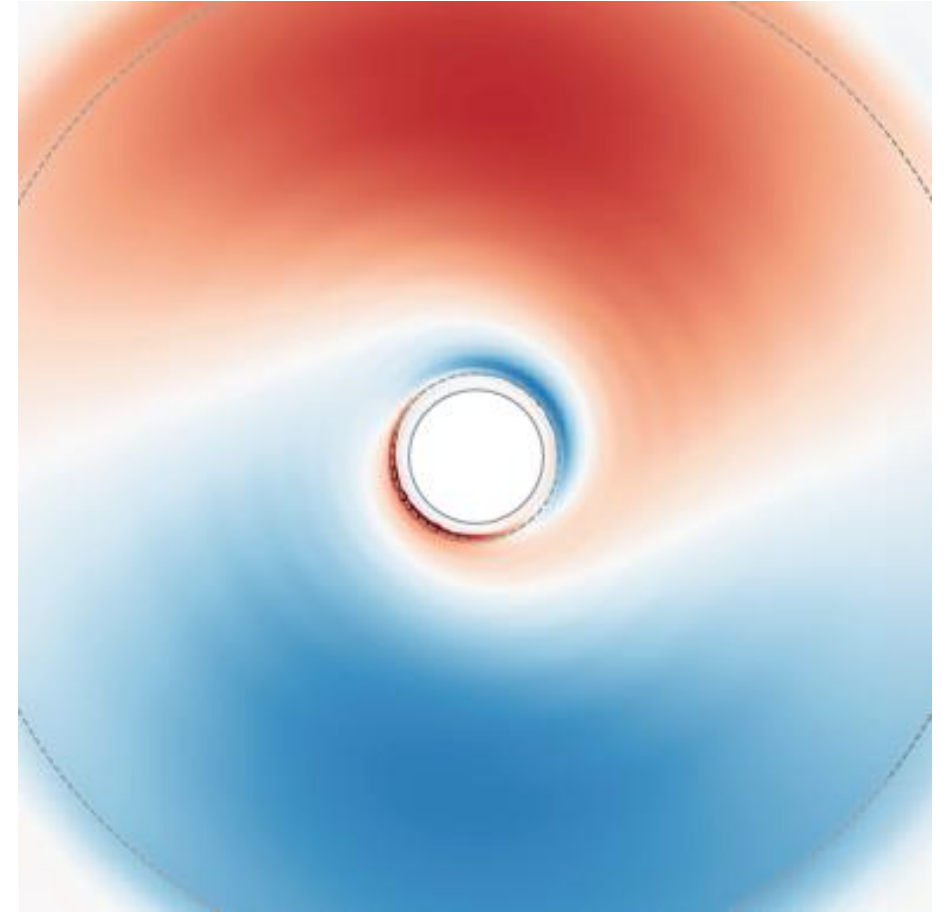


This is not a numerical artefact



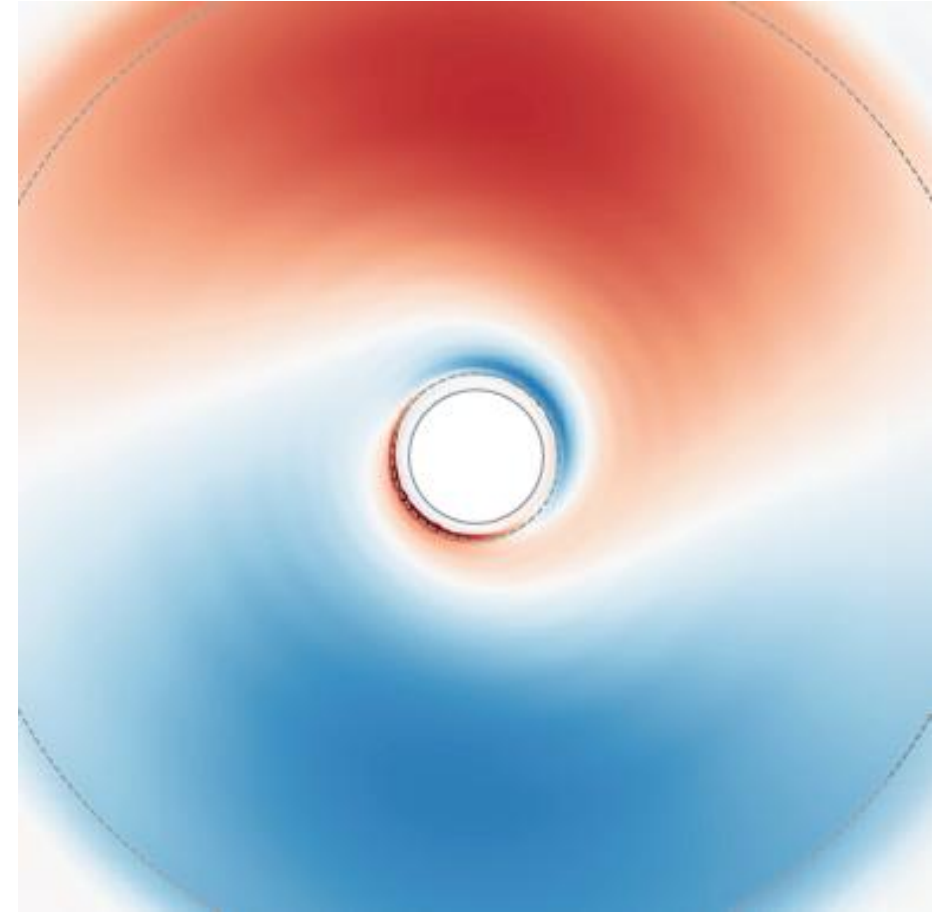
This ‘reflex instability’ (RI) is potentially important

- Creates spirals
 - Could explain some of the spirals observed by ALMA
 - Could make discs eccentric



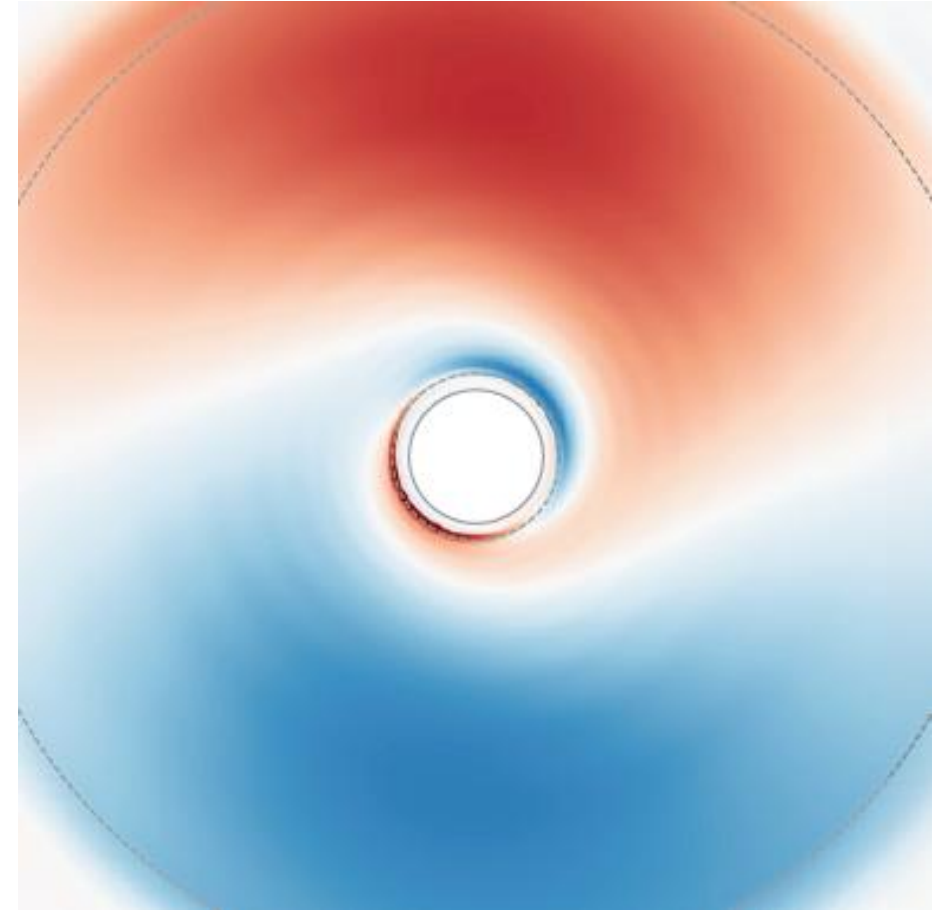
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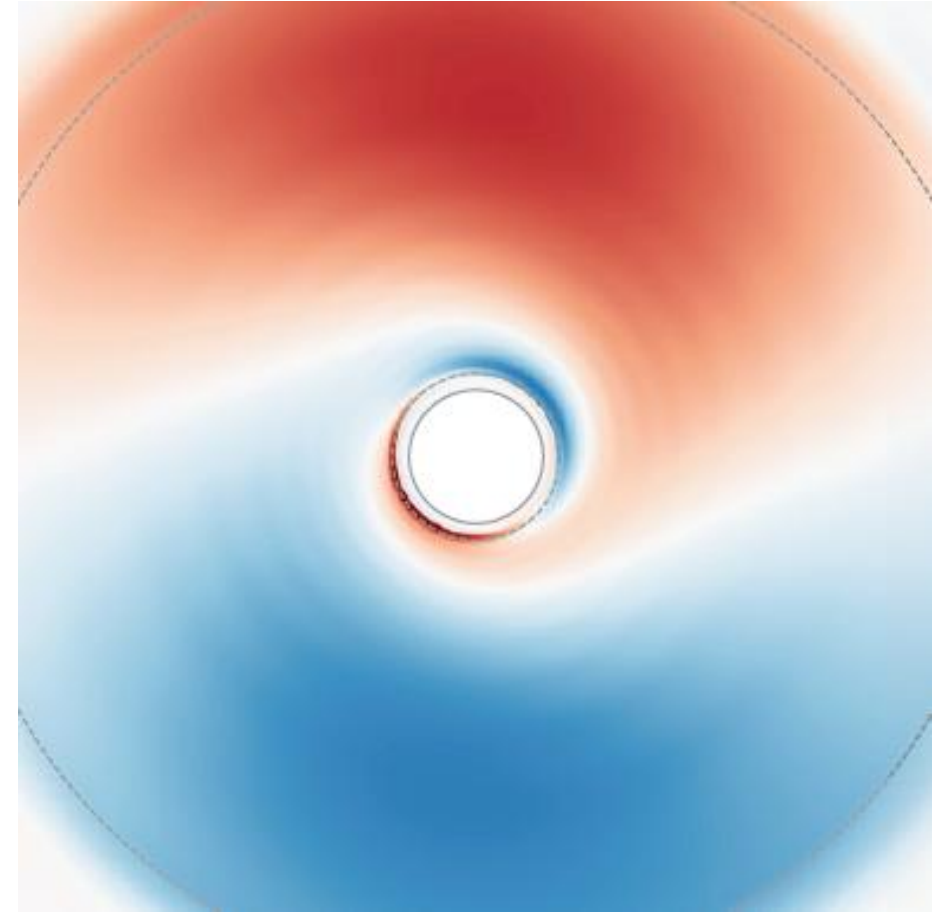
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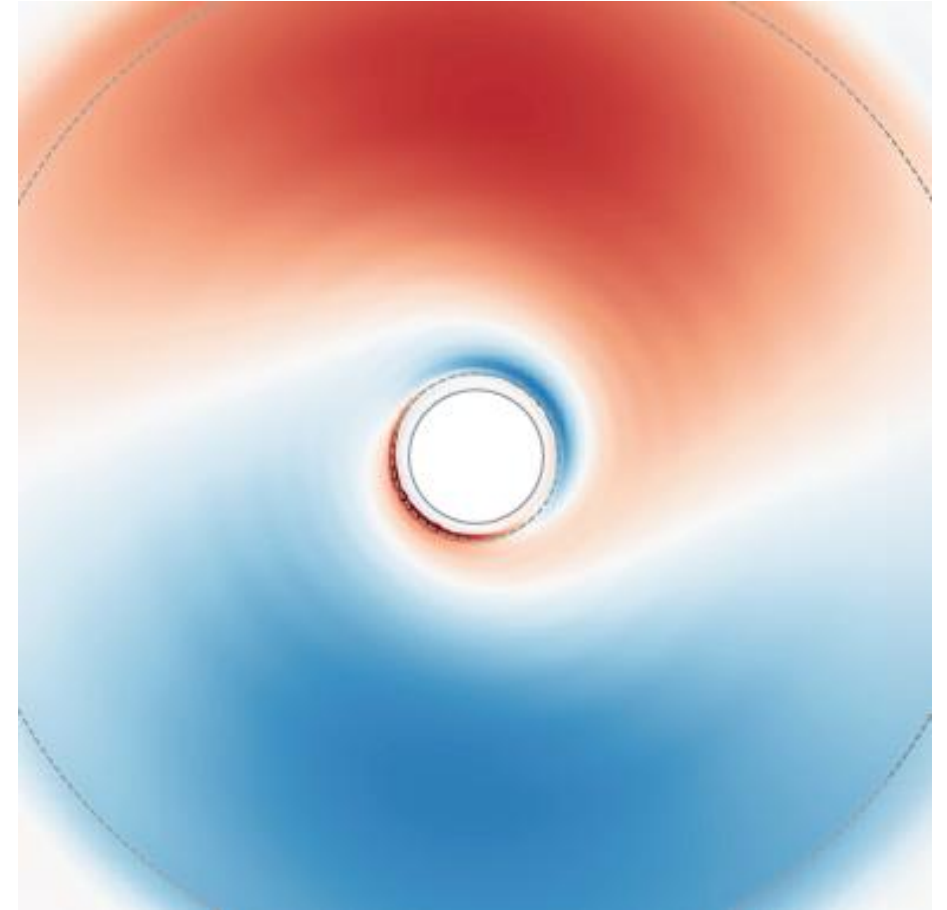
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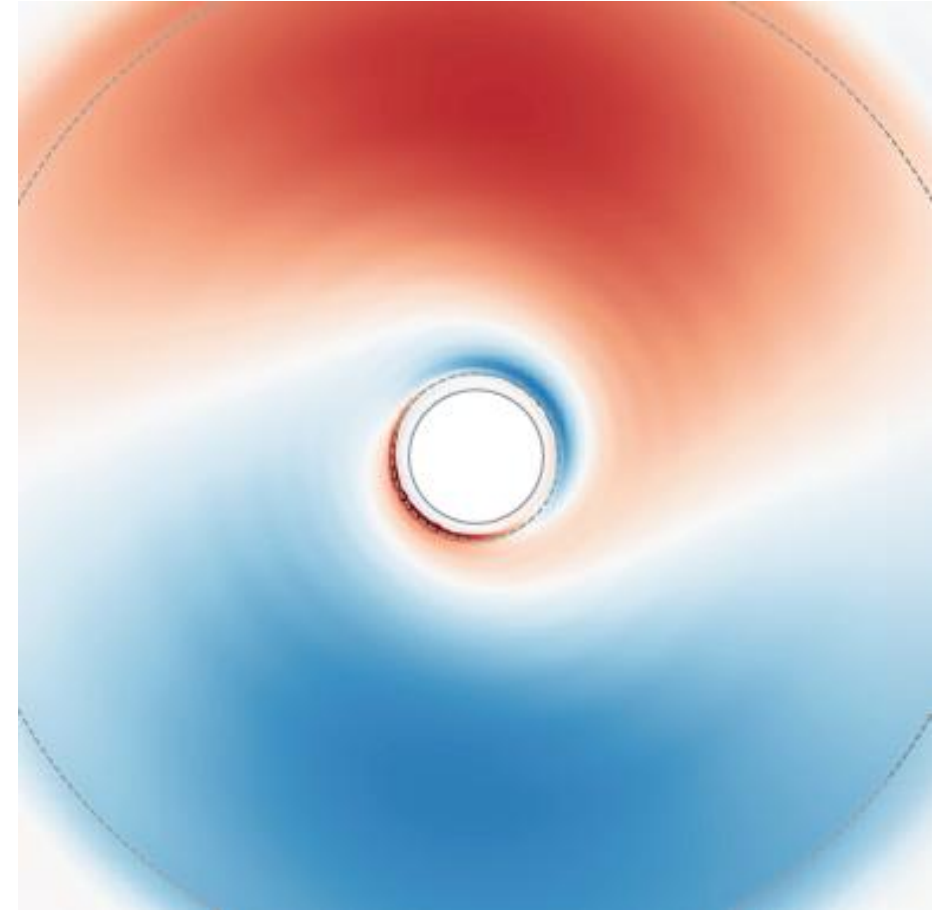
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BUT



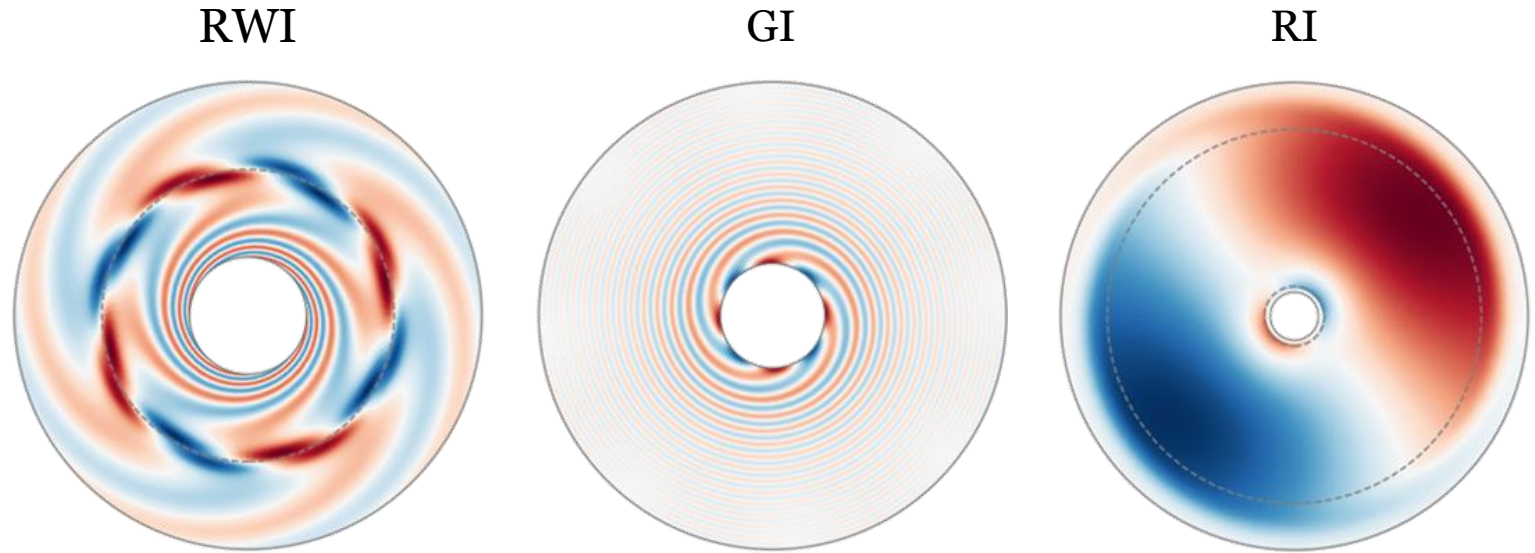
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- BUT**
- Disperses the disc in 10 kyr -100 kyr
 - There is observational evidence against that prediction
 - There must be a damping mechanism



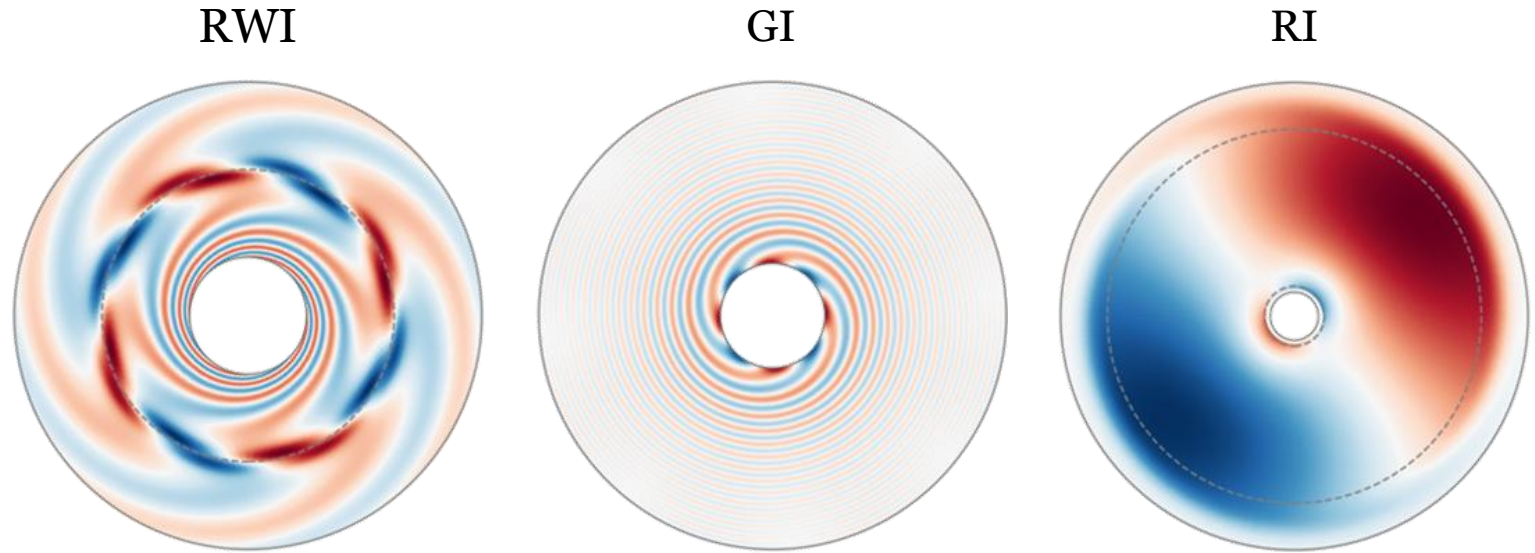
We're building LAADS, a code to study this instability (and many other)

- What works already:
 - 1.5D finite differences
 - Pade filters
 - Boundary conditions
 - Damping layers
 - GEVP Solver
 - Self-gravity module
 - Indirect term module



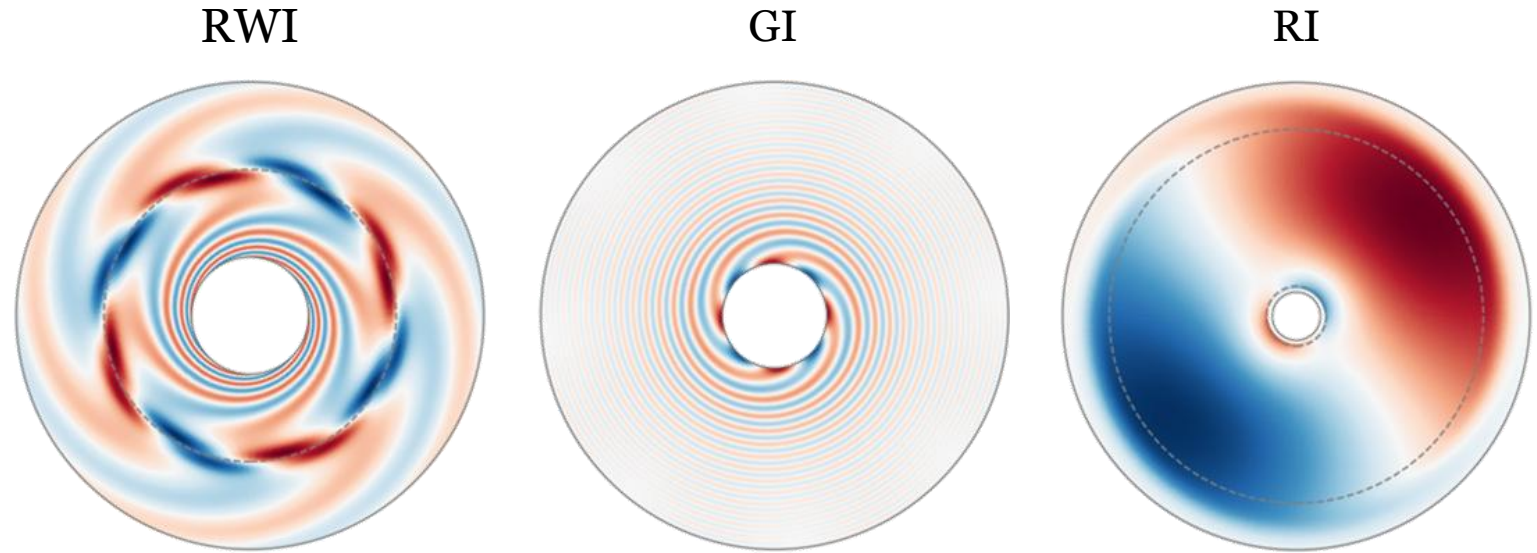
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- Computational cost:
 $\sim N_{\text{cell}}^{2.5} \mu\text{s}$



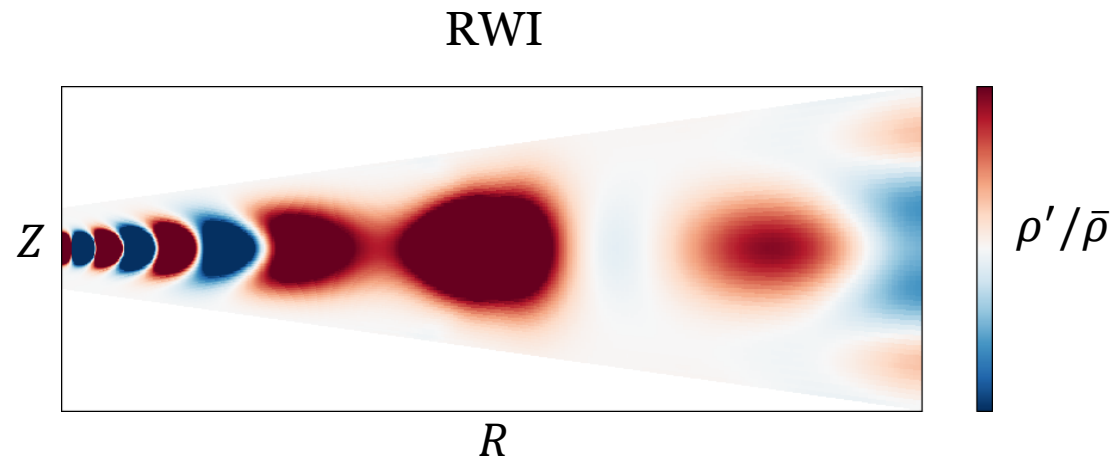
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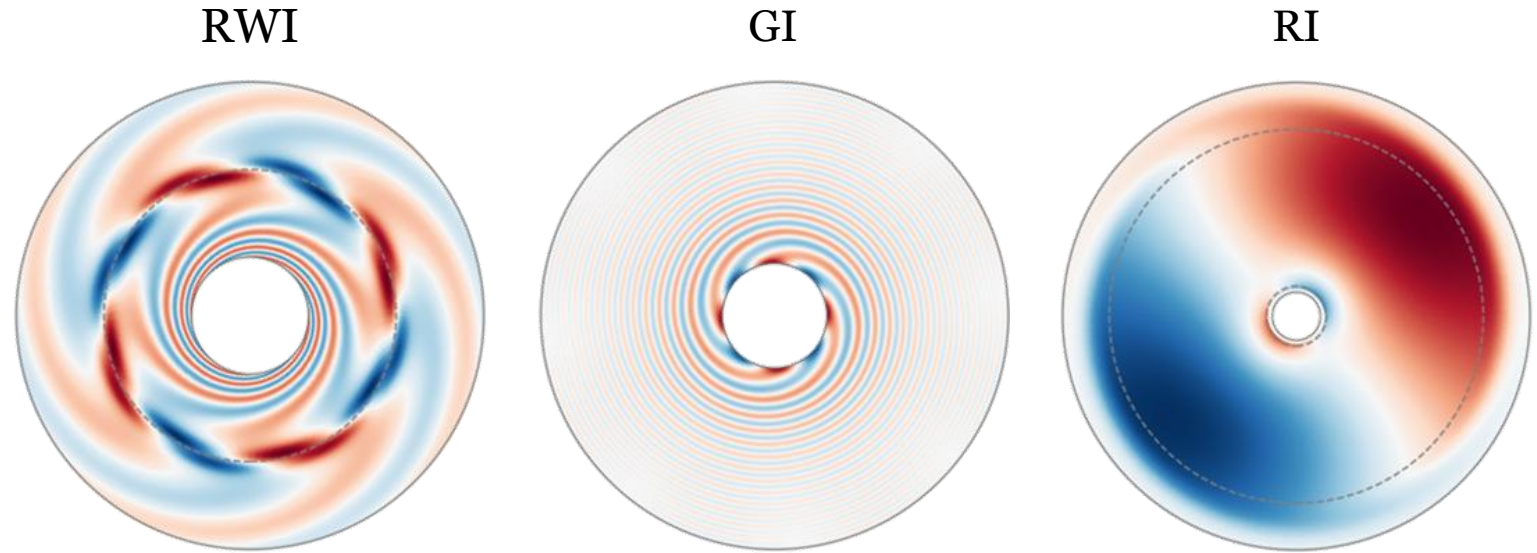
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 - 2.5D finite differences
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 - RWI, VSI, ...



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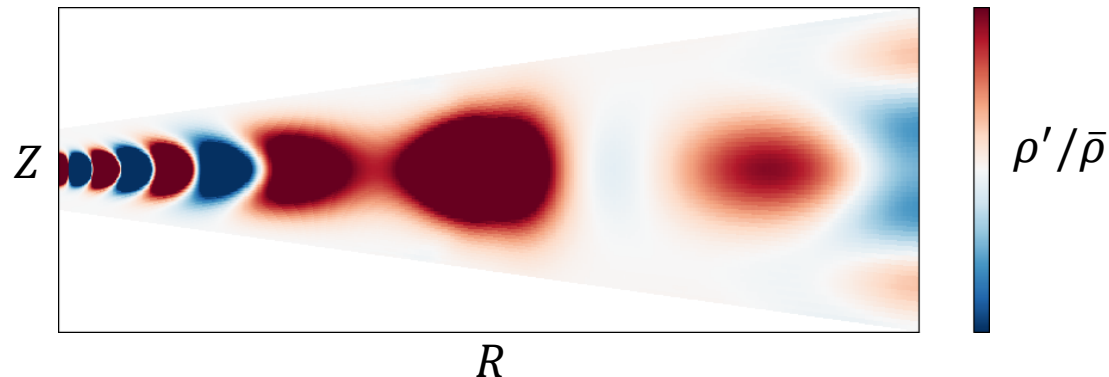
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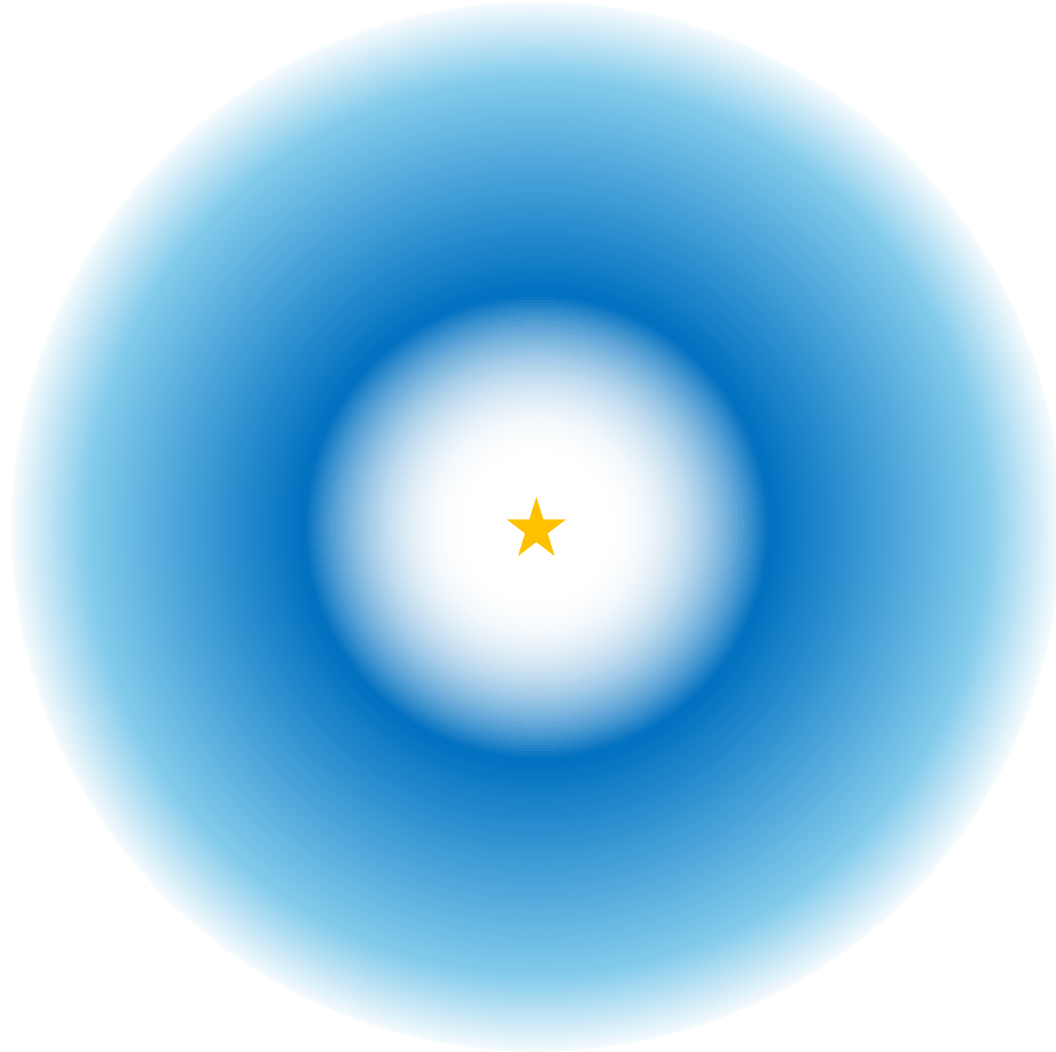
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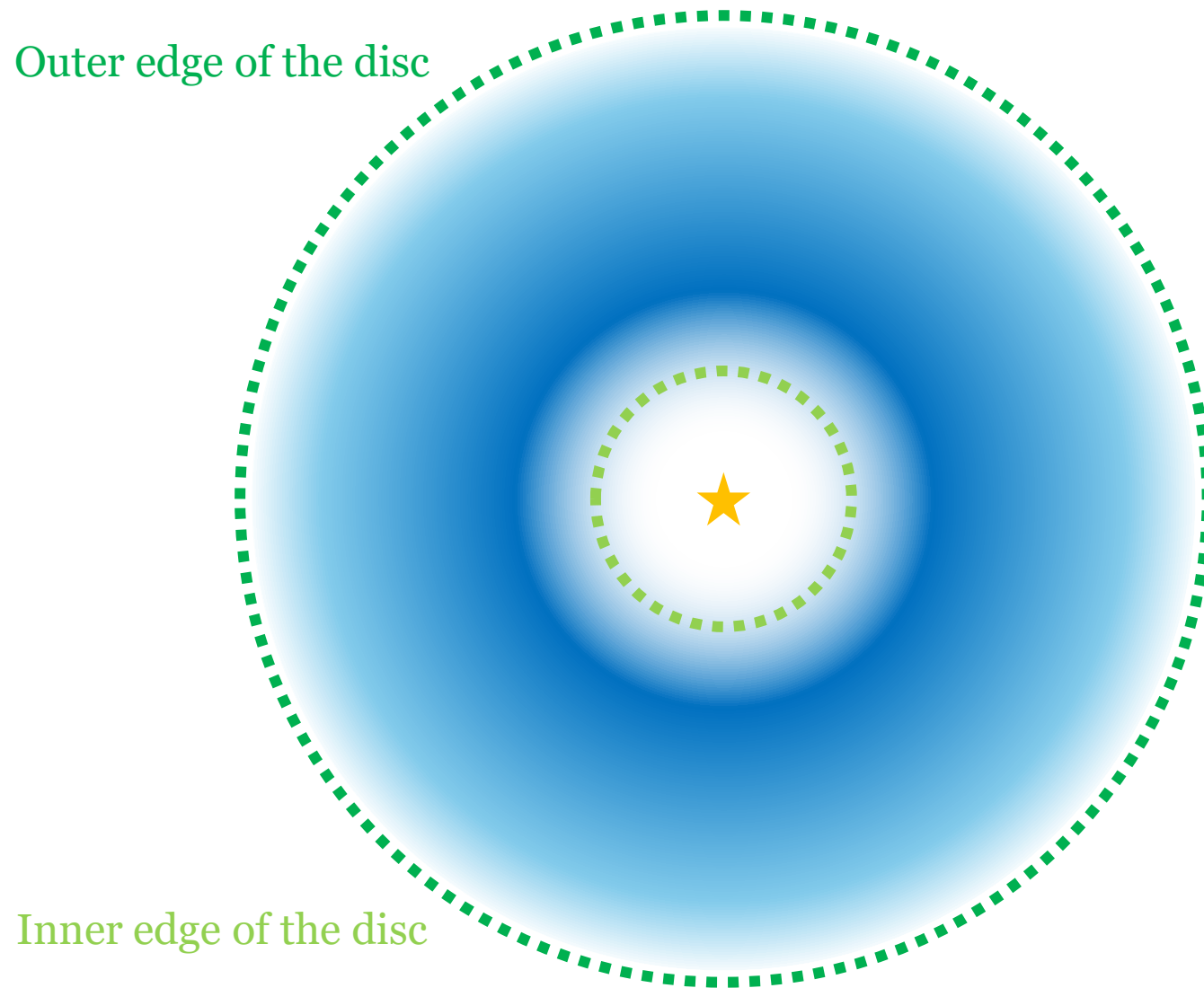
This is Enzo Dupont's master's thesis



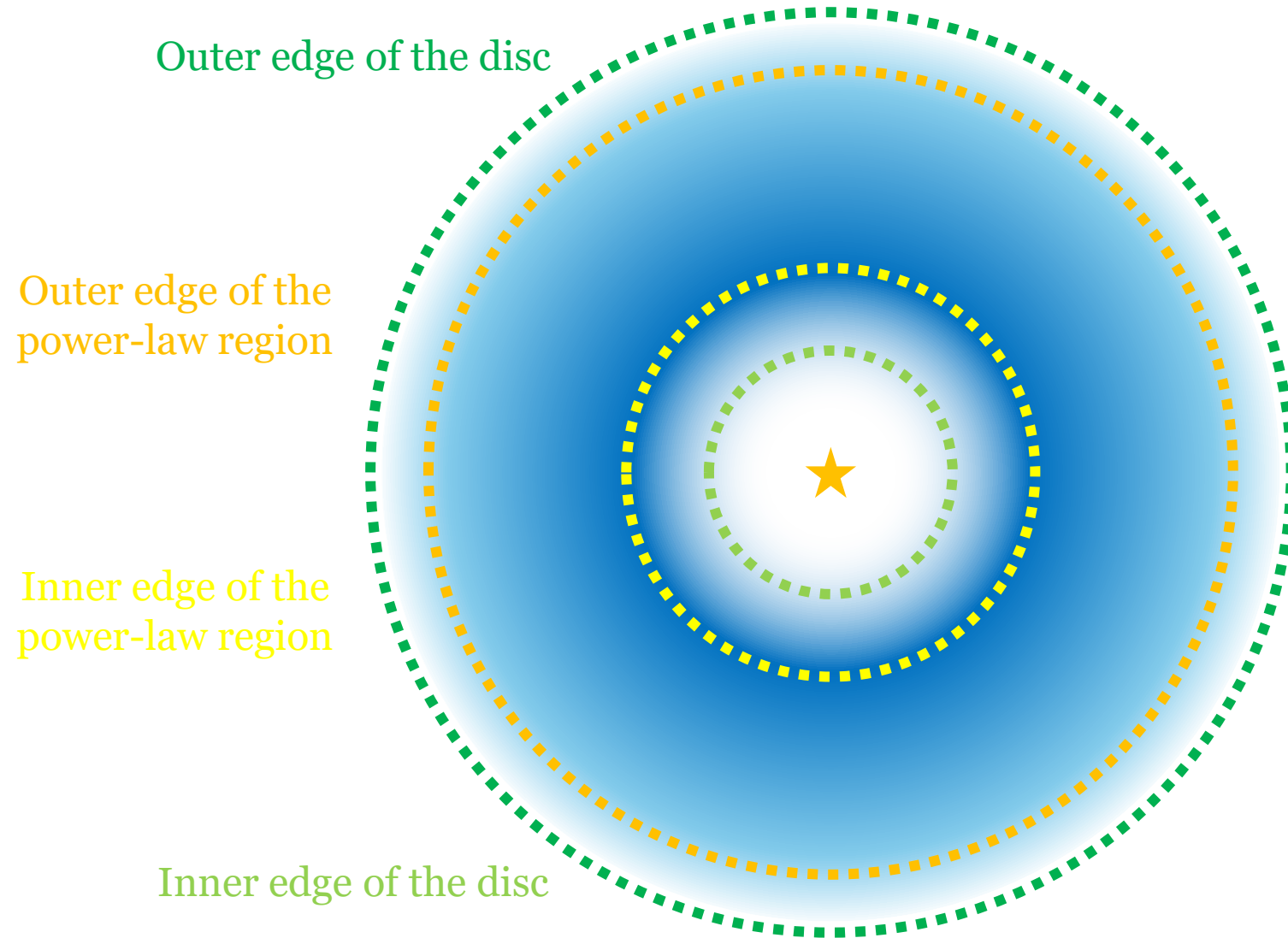
No one ever models the whole disc (except those pesky SPH people)



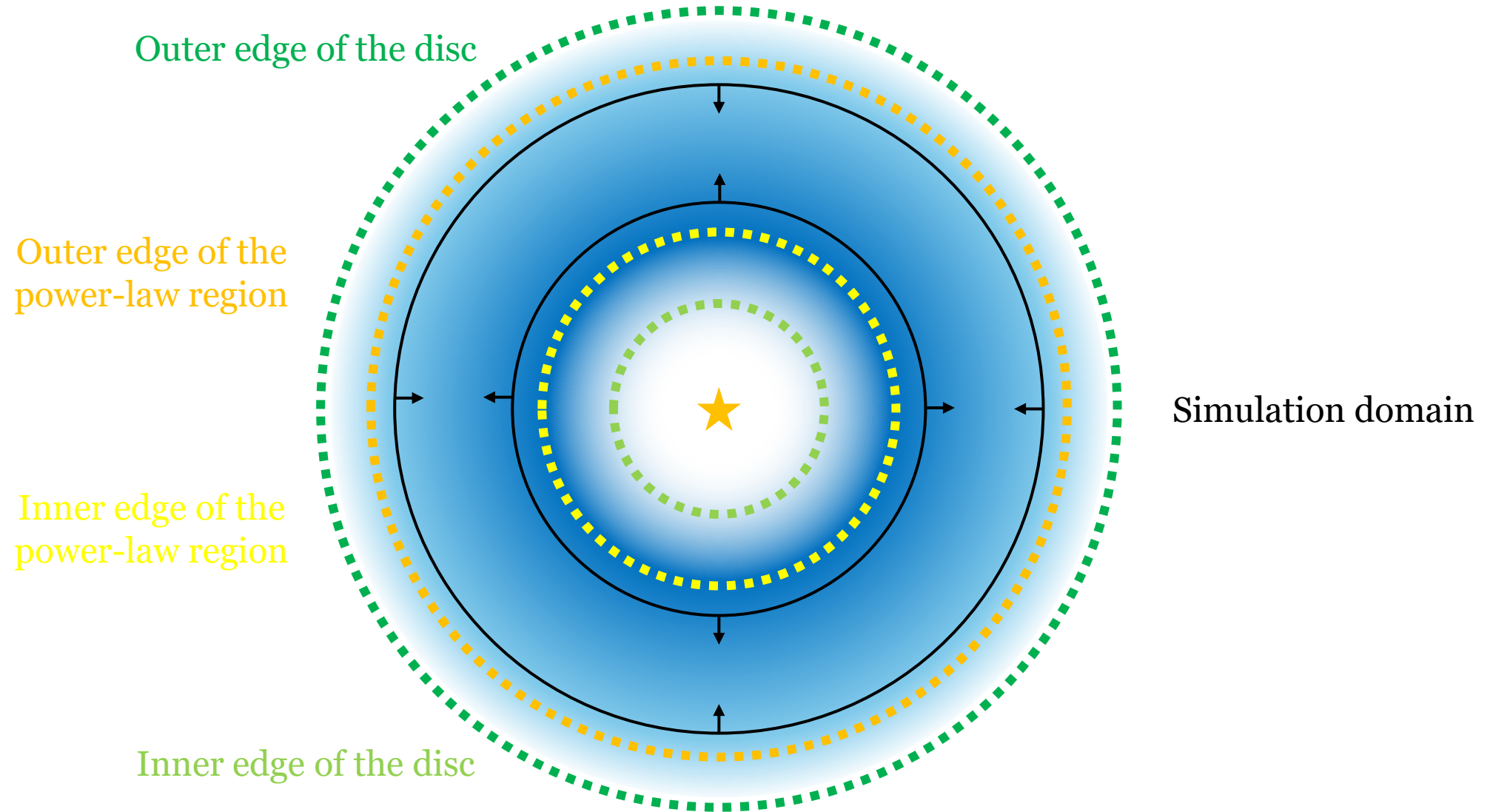
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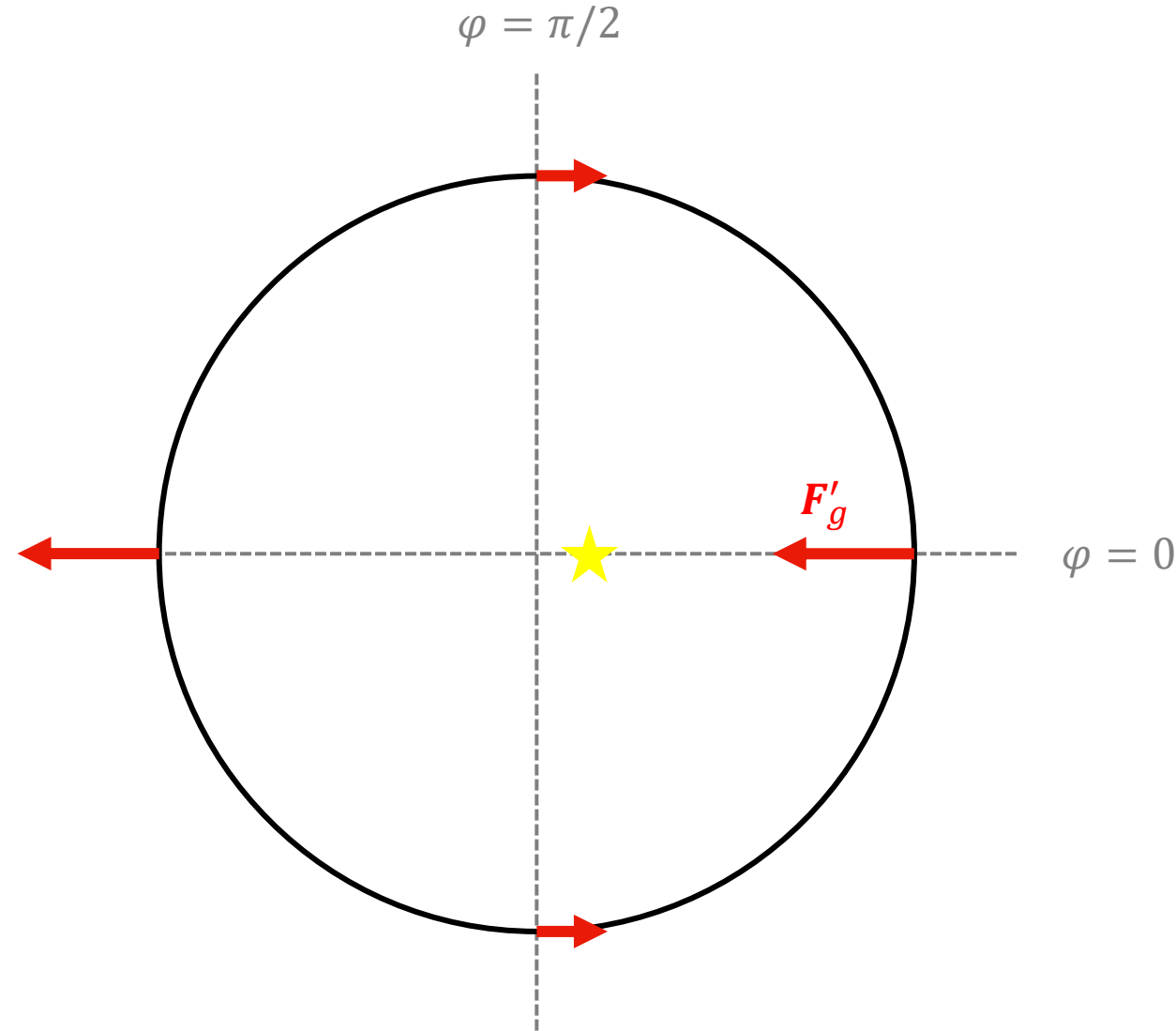


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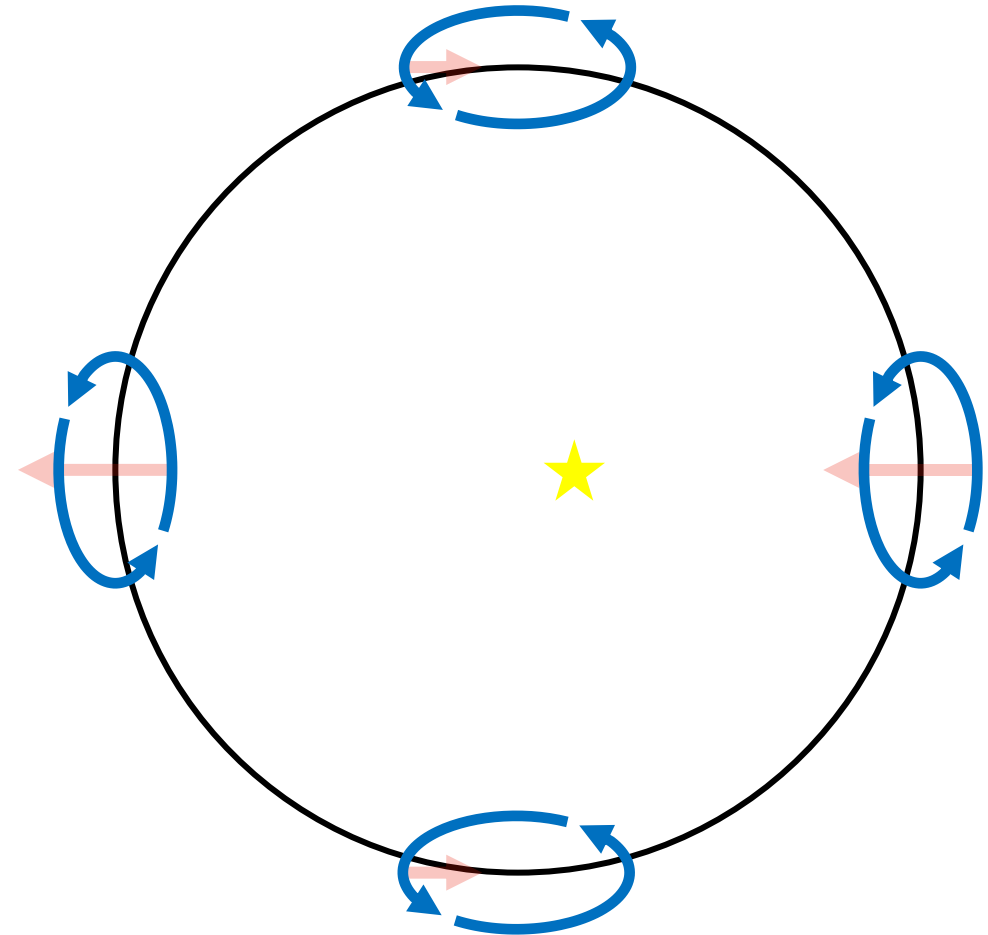
This restriction triggers a first class of reflex modes: the ‘bracket’ modes

- Let us imagine that the star is off-centre
→ Its gravitational pull is perturbed



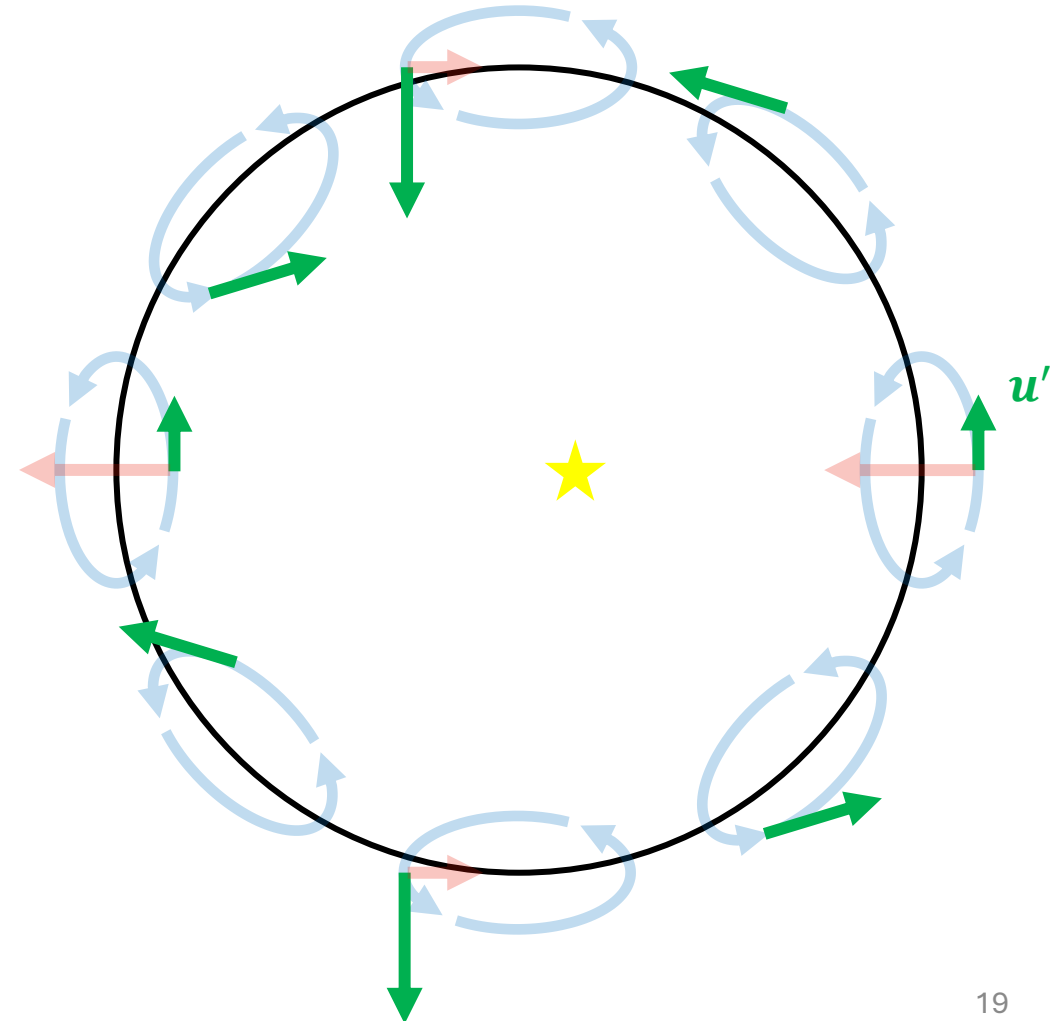
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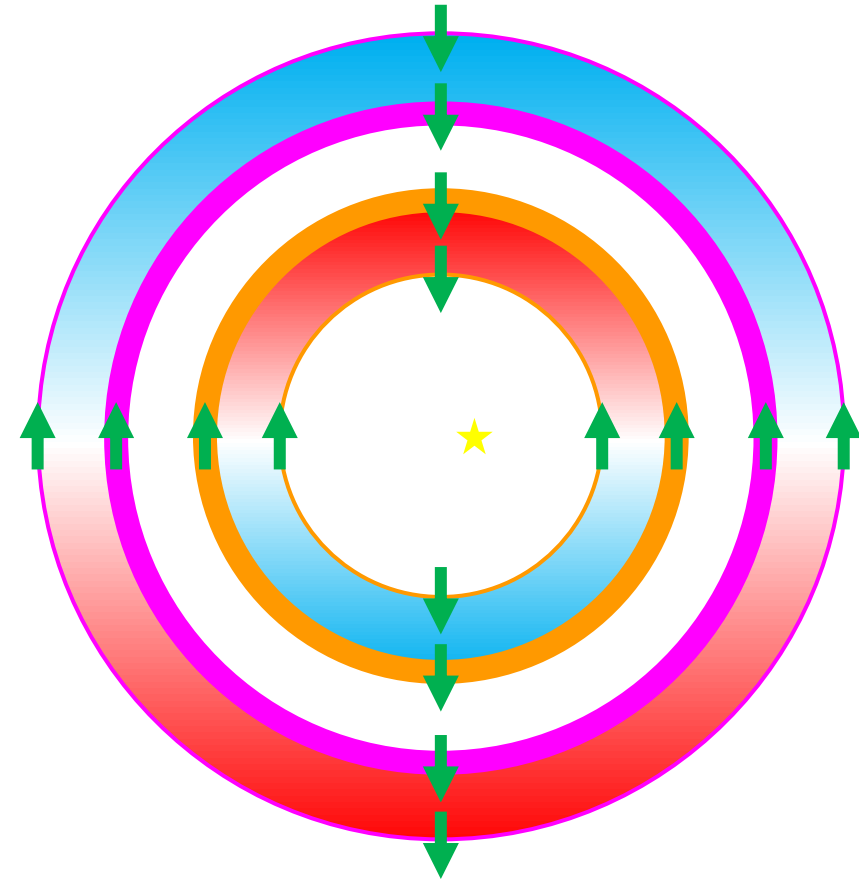
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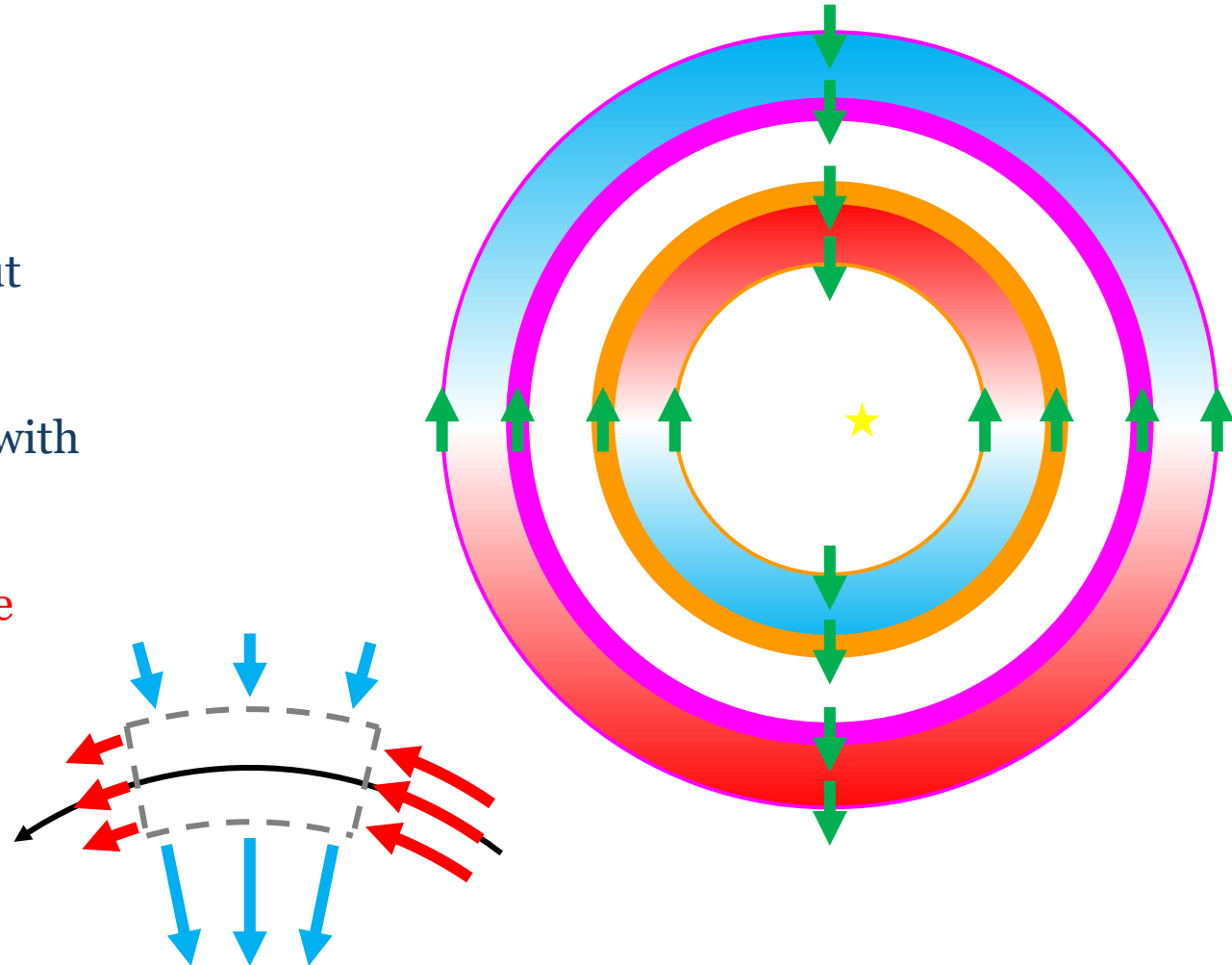
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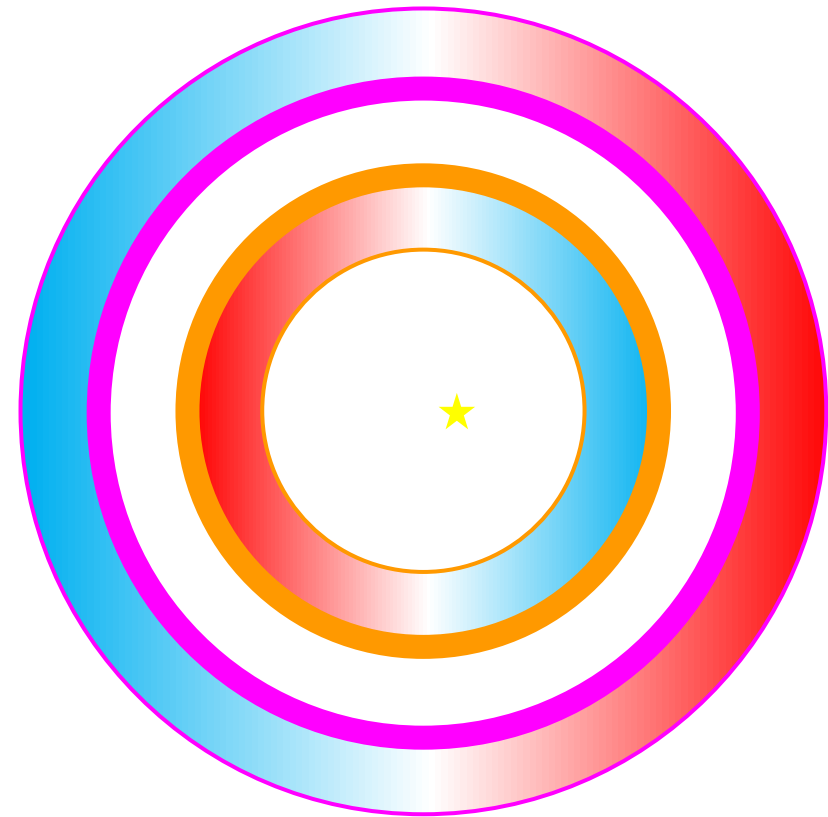
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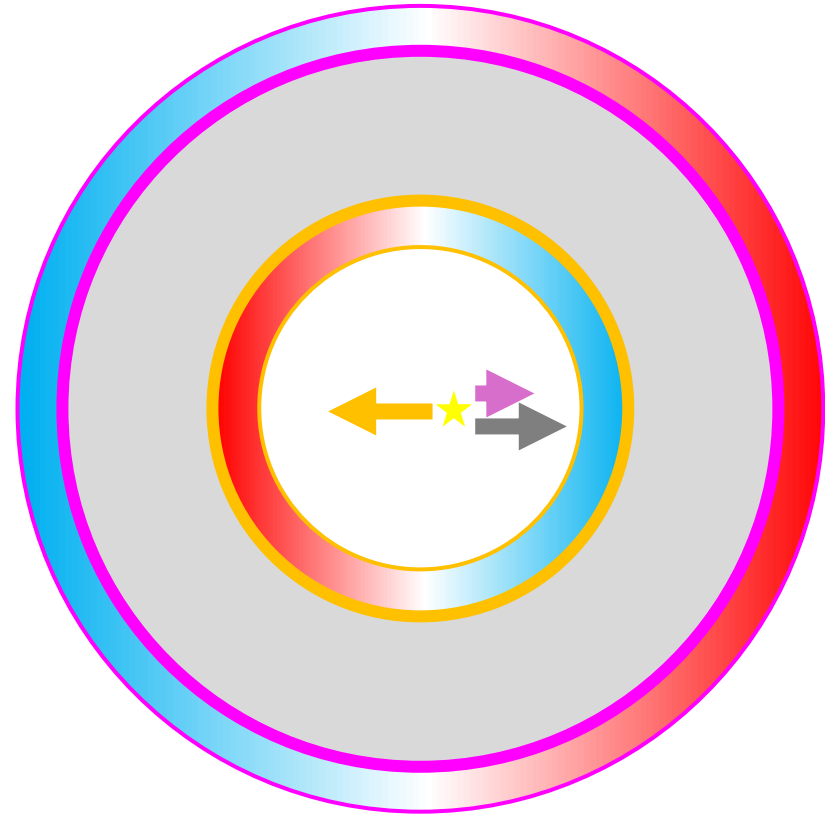
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- The density perturbation changes sign between the inner disc and the outer disc



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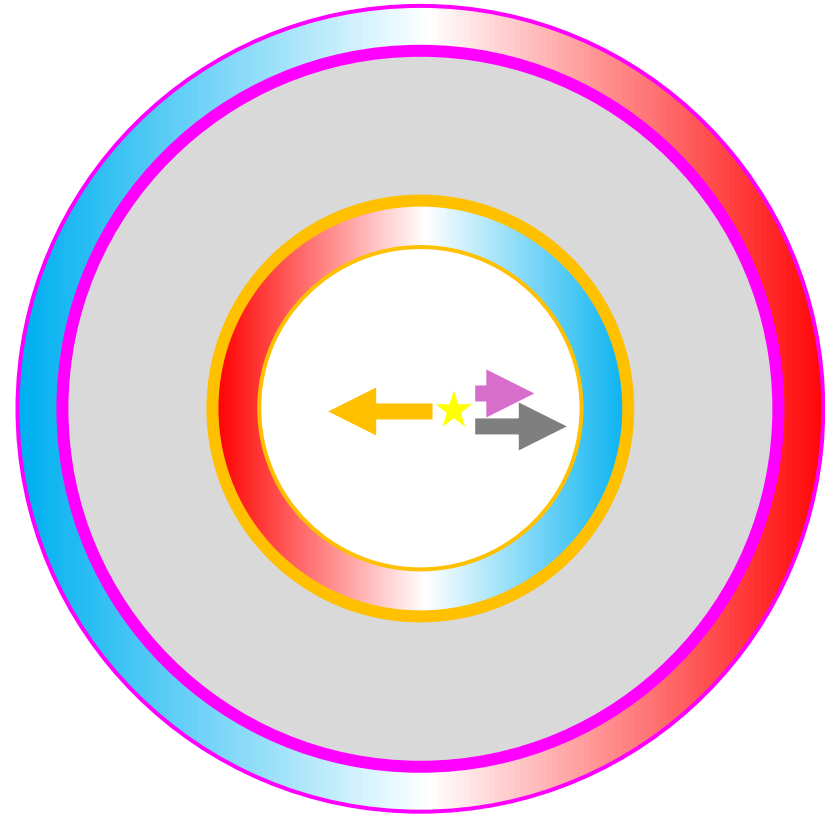
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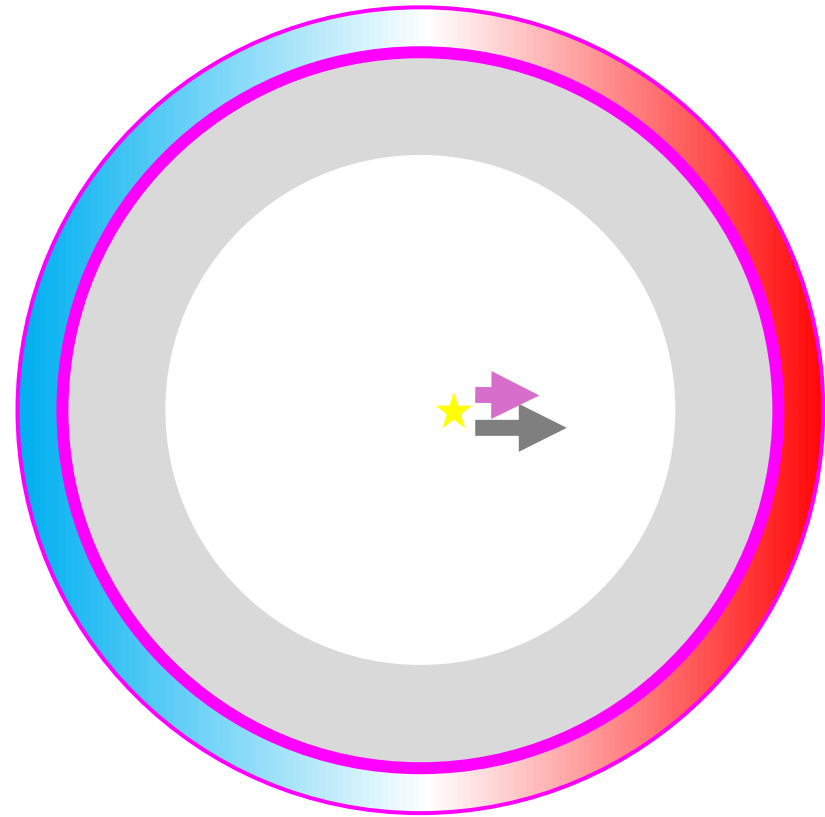


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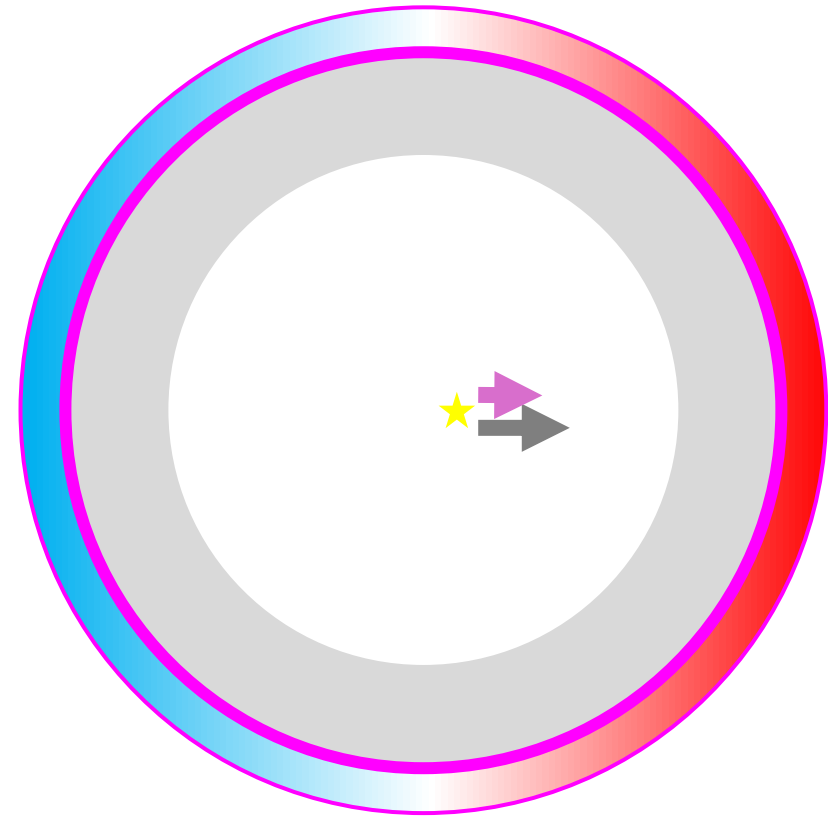


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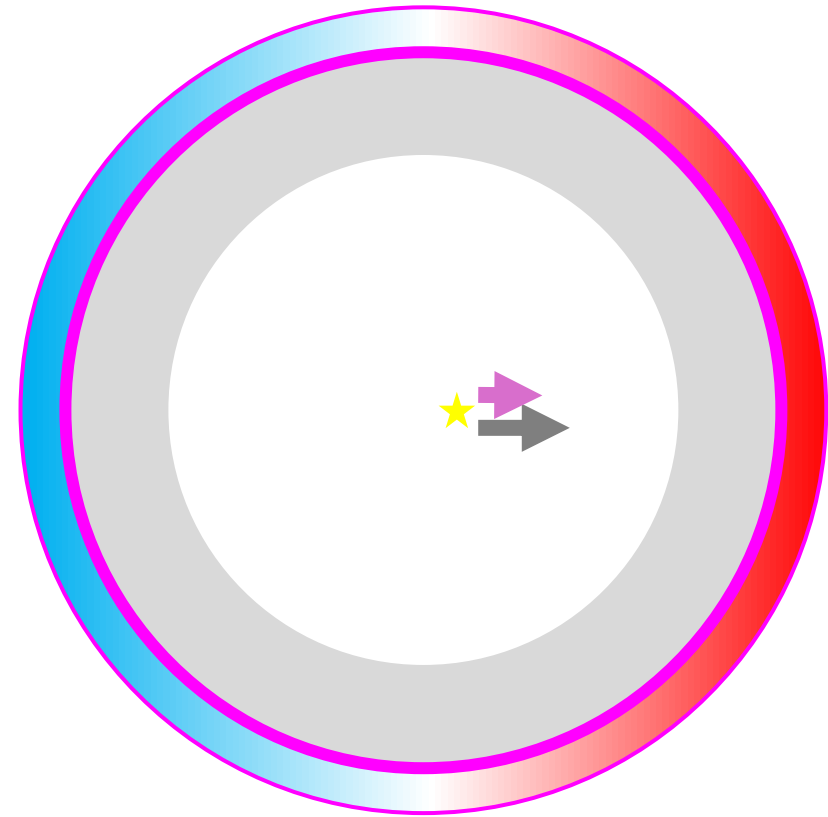
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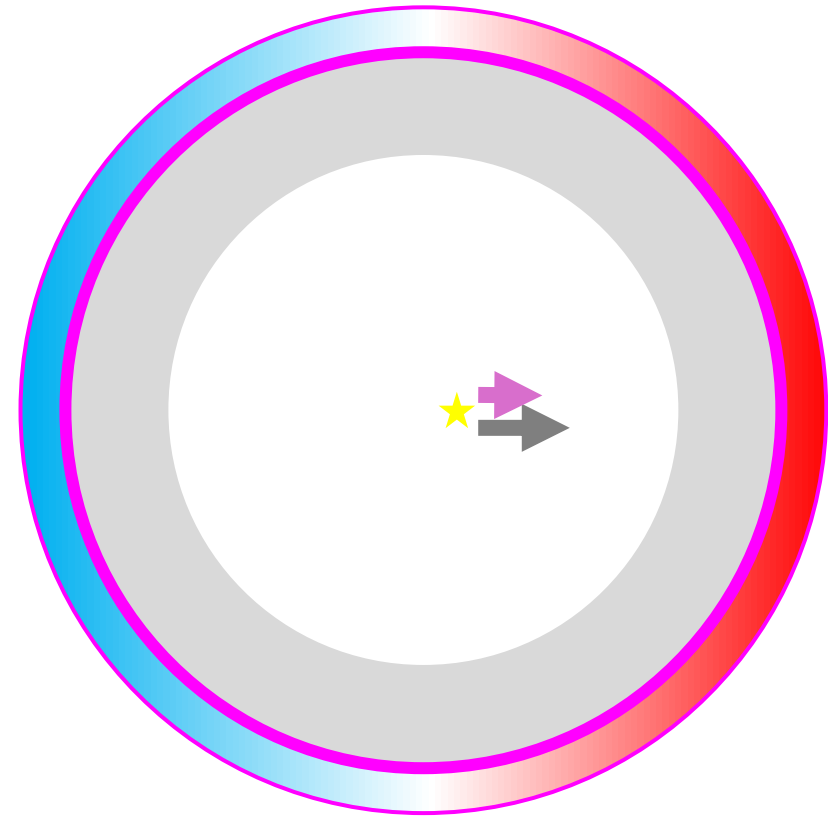
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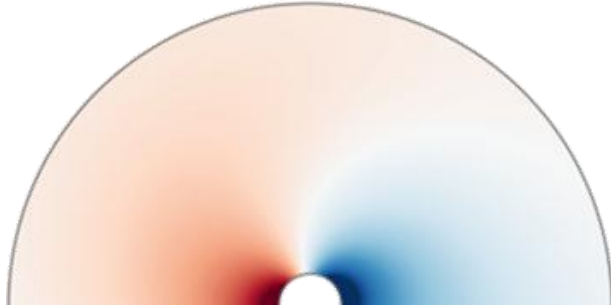
The star goes further right, instability

But this is clearly an artifact



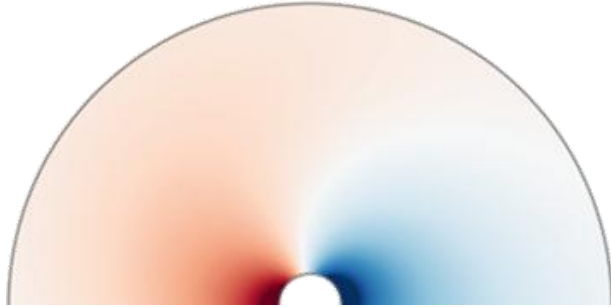
It turns out that there are (at least) nine classes of ‘reflex’ modes

Bracket modes

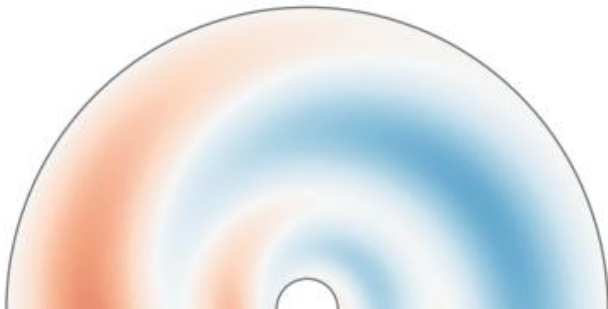


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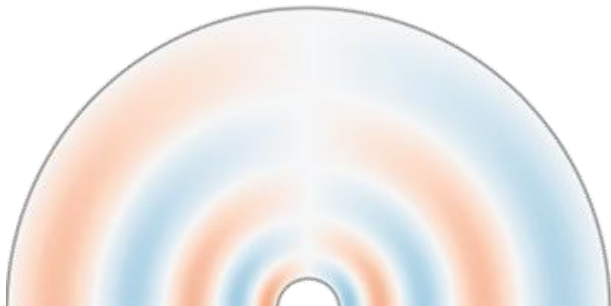


Standard cavity modes

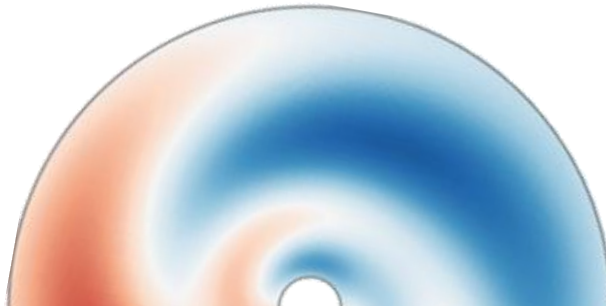


(Adding pressure)

Resonant cavity modes

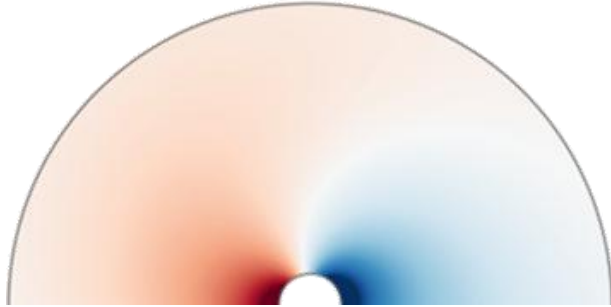


Collisional modes ($\times 2$)

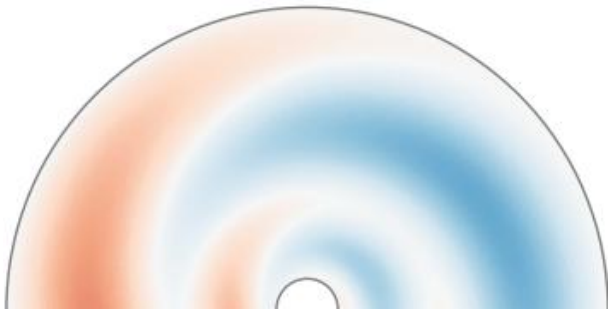


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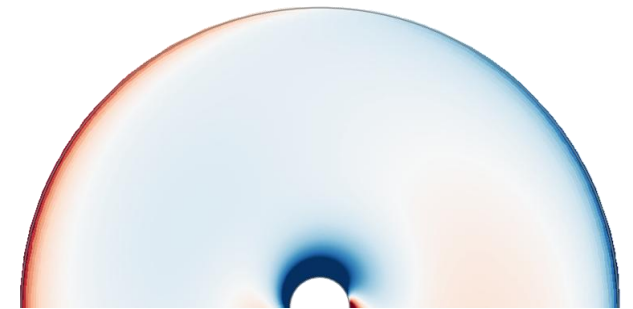
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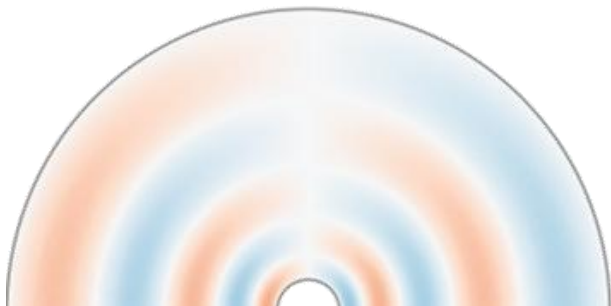
Standard cavity modes



Evanescent modes

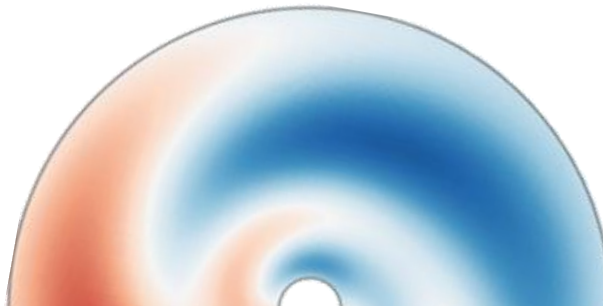


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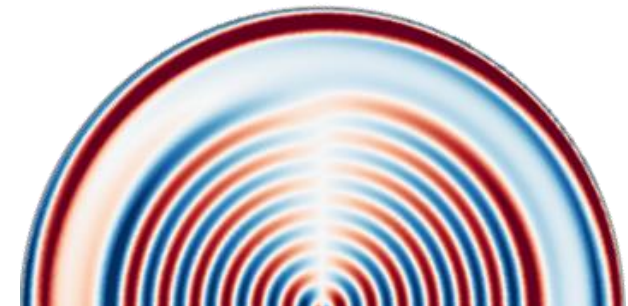


(Adding self-gravity)

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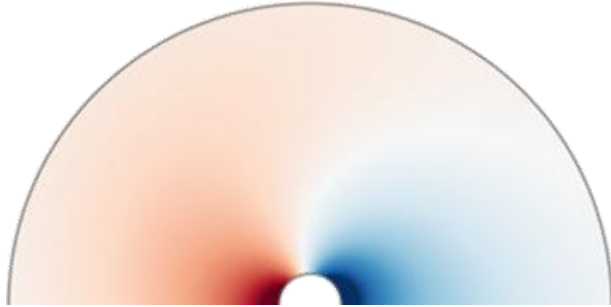


SLING modes

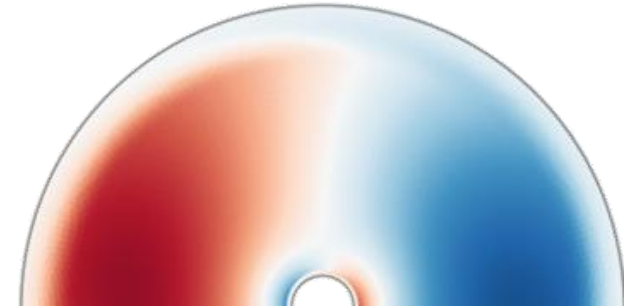


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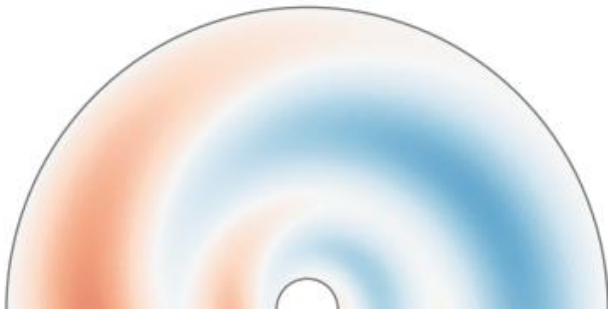
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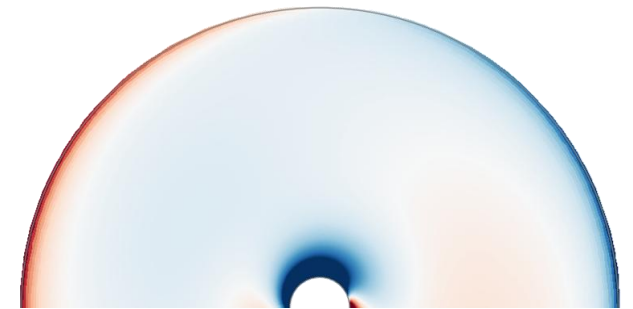
Overdamped modes



Standard cavity modes

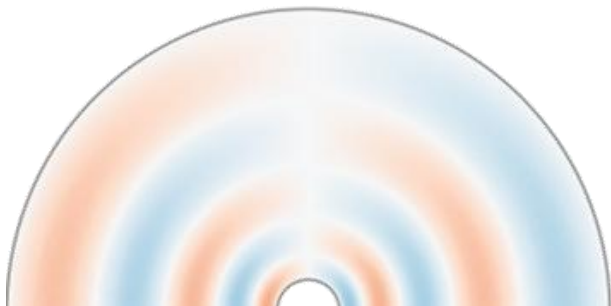


Evanescent modes

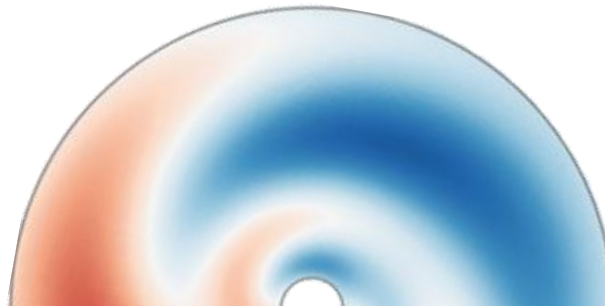


(Adding damping layer)

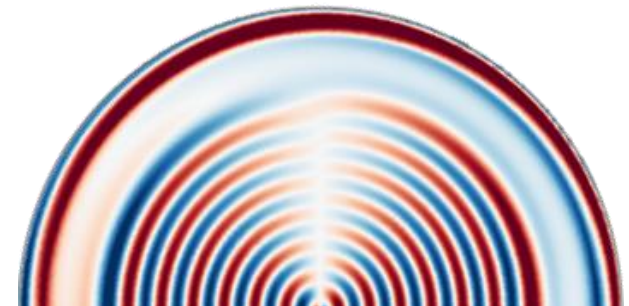
Resonant cavity modes



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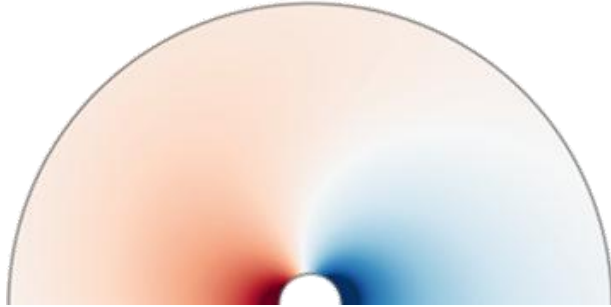


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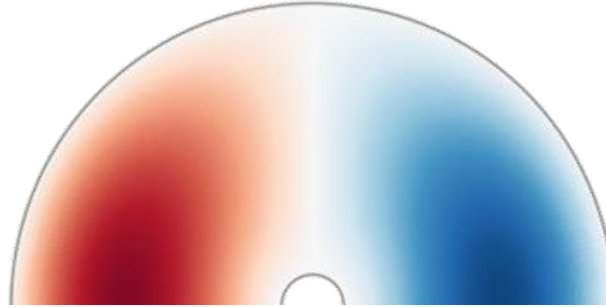


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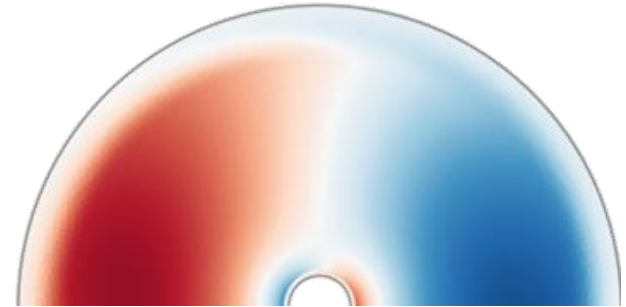
Bracket modes



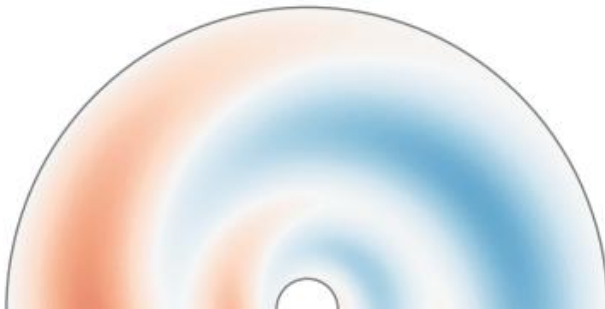
Those guys



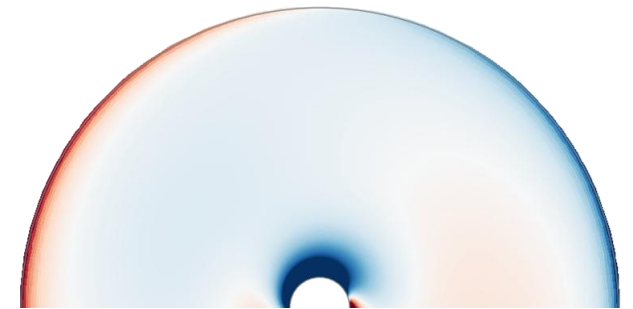
Overdamped modes



Standard cavity modes

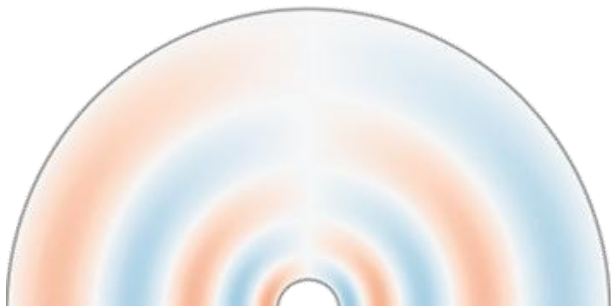


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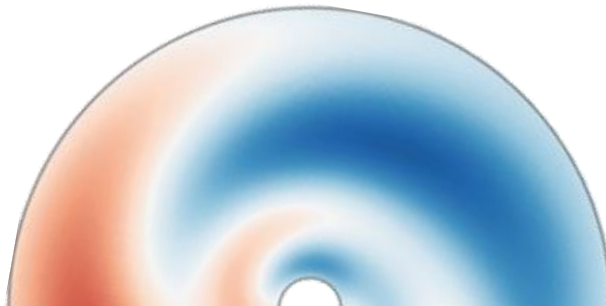


(We don't understand
this one yet, so we
haven't named it)

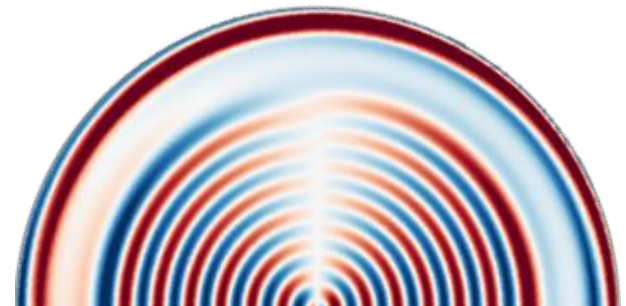
Resonant cavity modes



Collisional modes ($\times 2$)

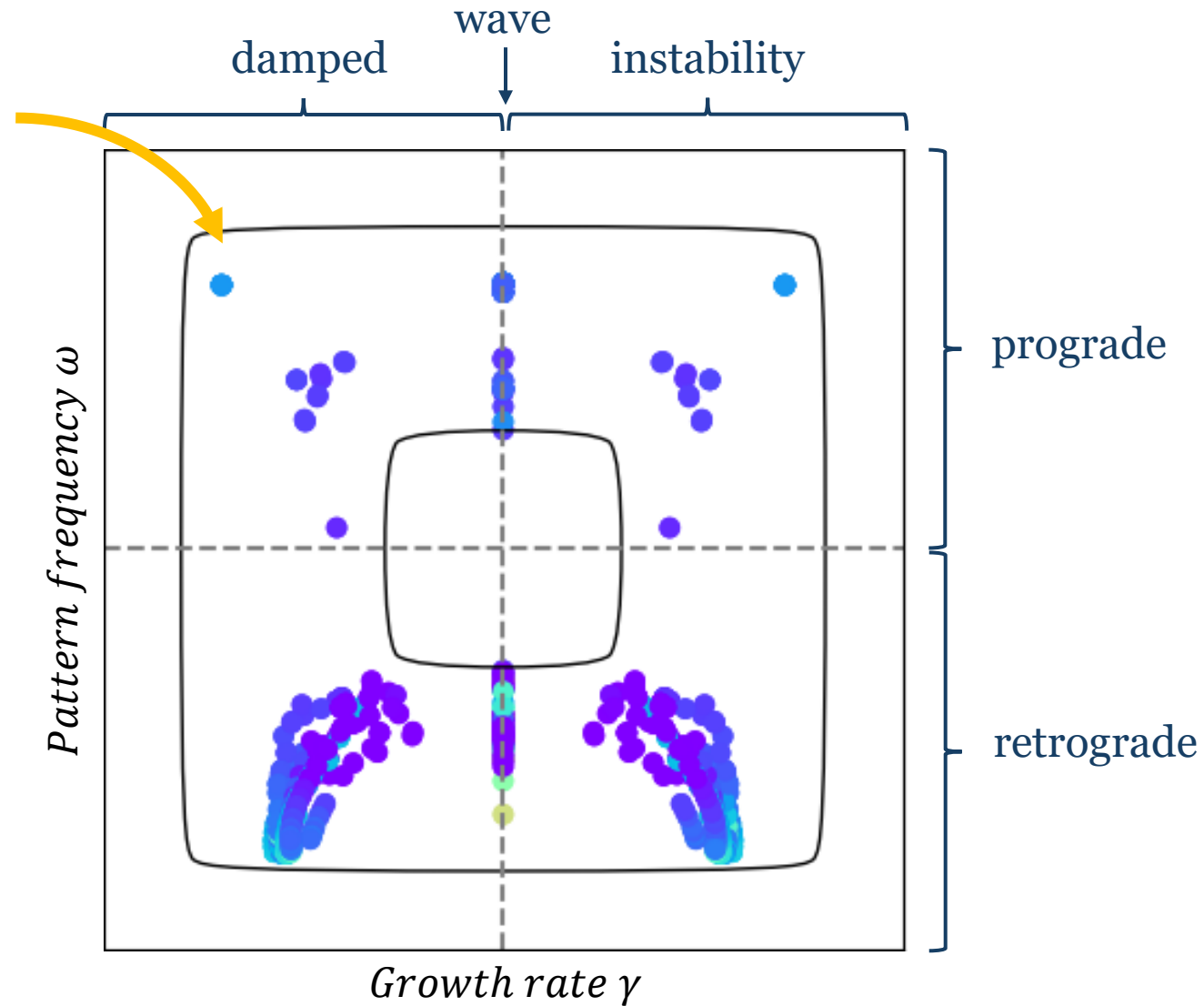


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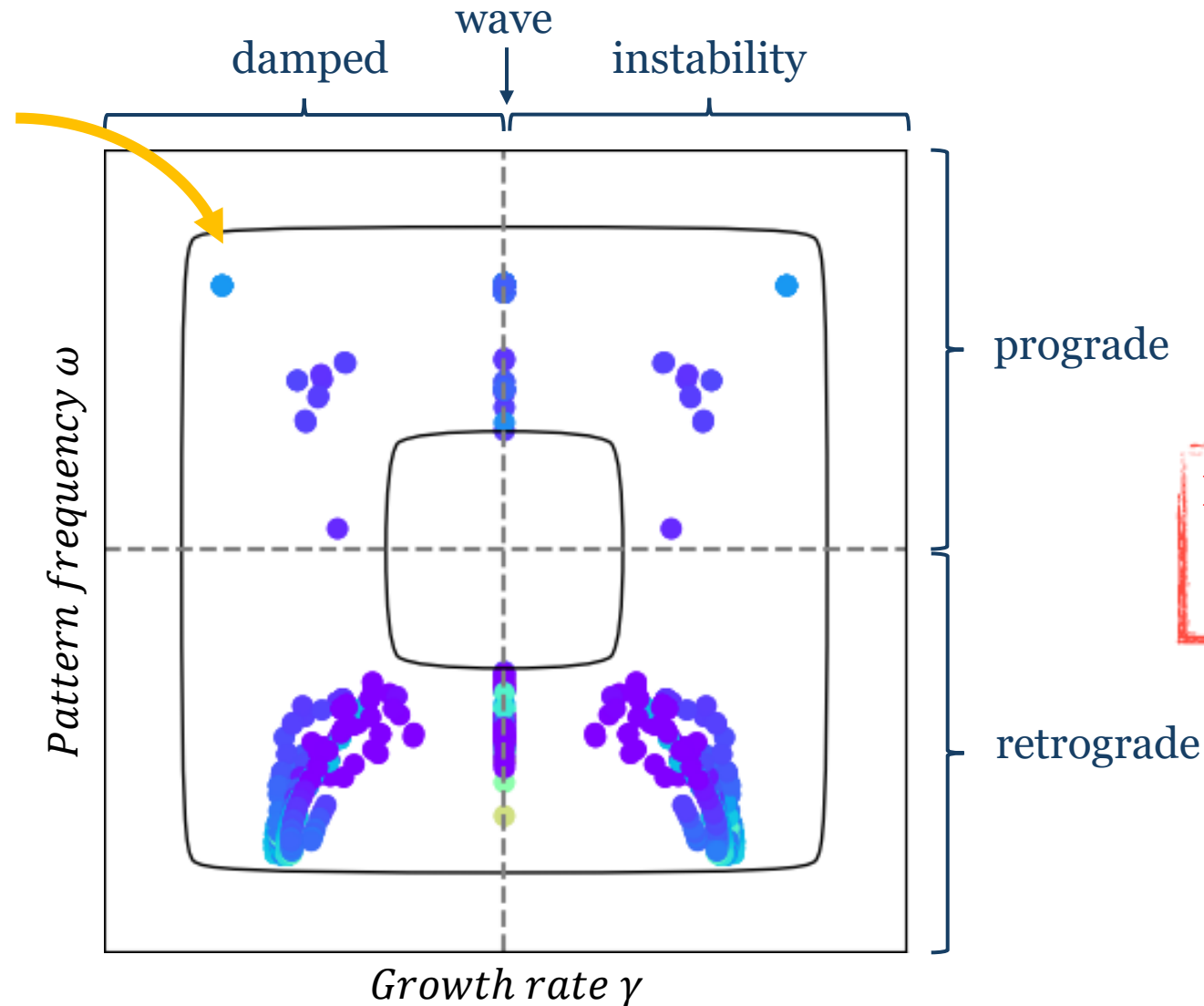
Bad news: reflex modes are very sensitive to boundary conditions

Complex frequencies of the cavity modes for 120 sets of boundary conditions



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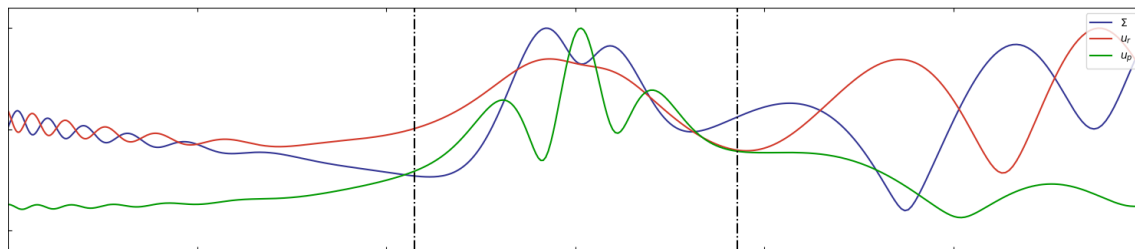


Very bad news: we can't get rid of those modes with damping layers

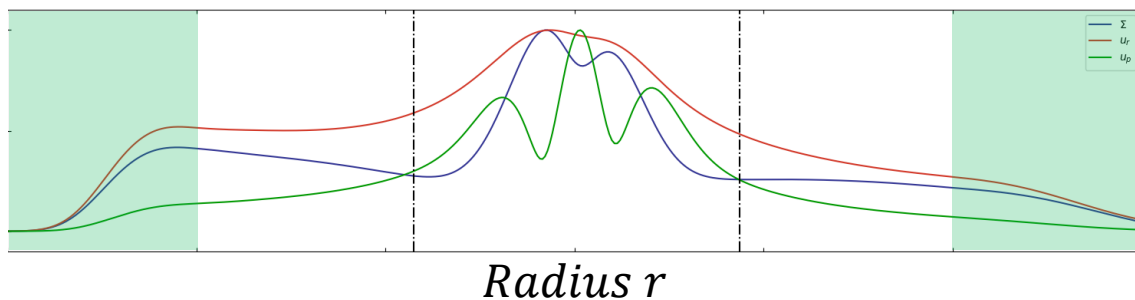
Damping layer = an artificial damping term added to each equation near the boundaries ($d_t f' = -\mathbf{d}(\mathbf{r}) f'$)

It works for the RWI

Without damping layers



With damping layers

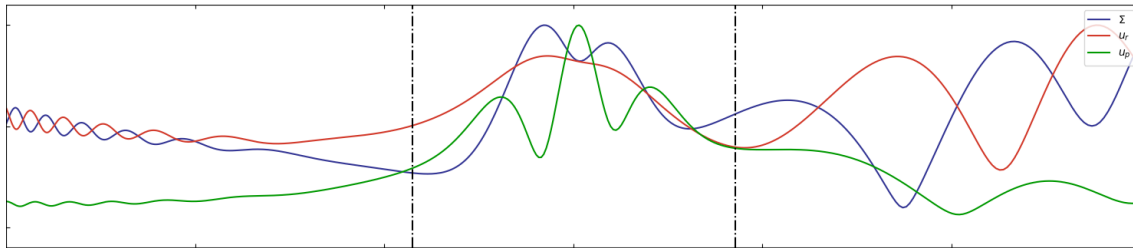


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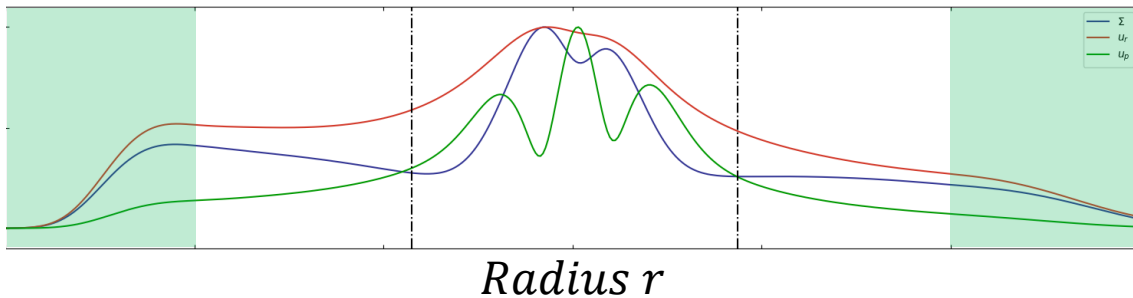
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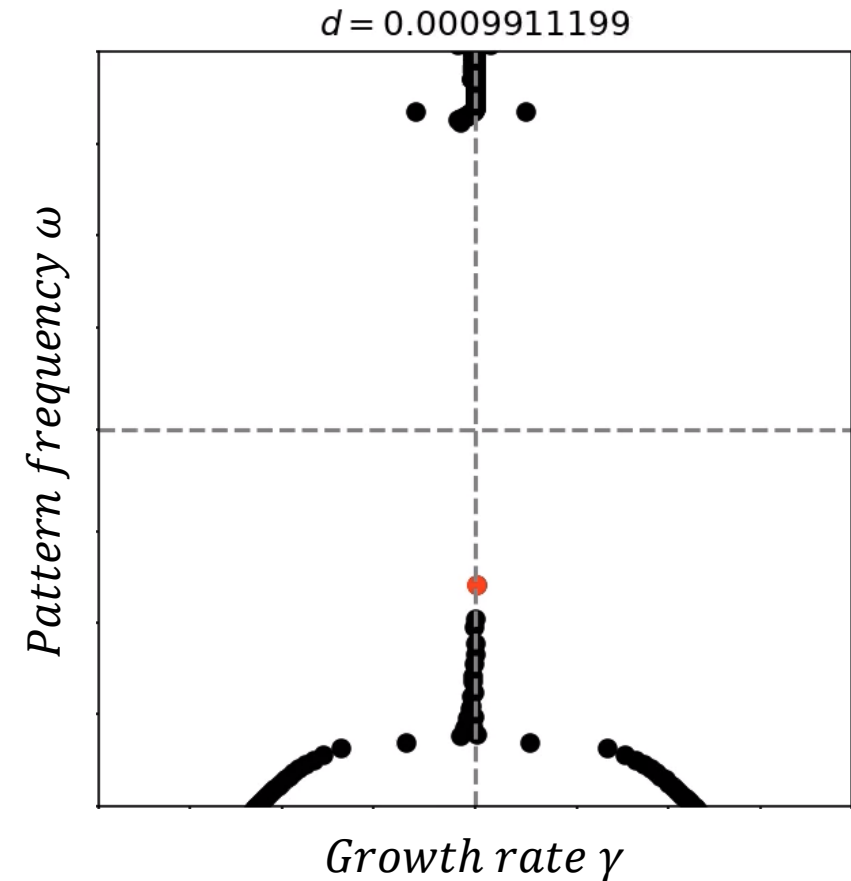
Without damping layers



With damping layers



But not for all cavity modes

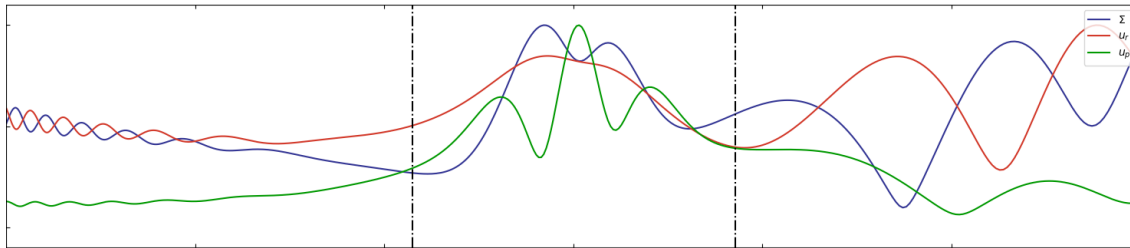


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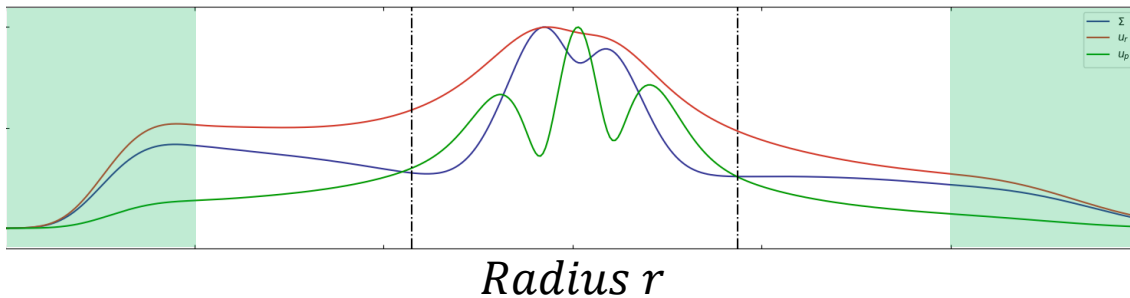
Damping layer = an artificial damping term added to each equation near the boundaries ($d_t f' = -d(r) f'$)

It works for the RWI

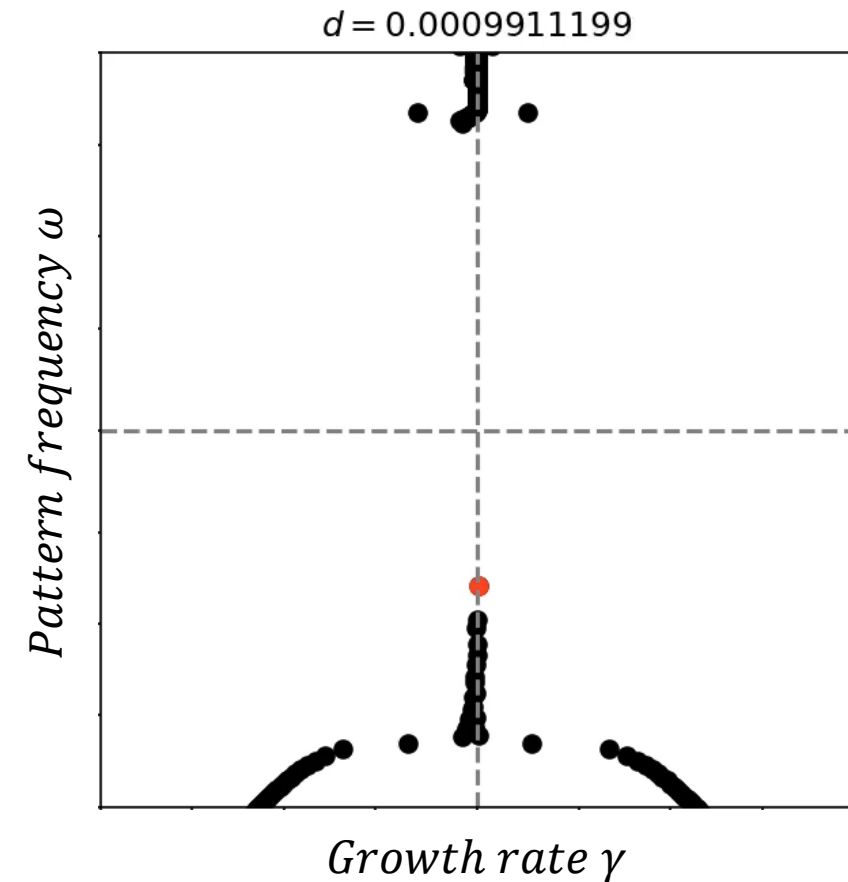
Without damping layers



With damping layers



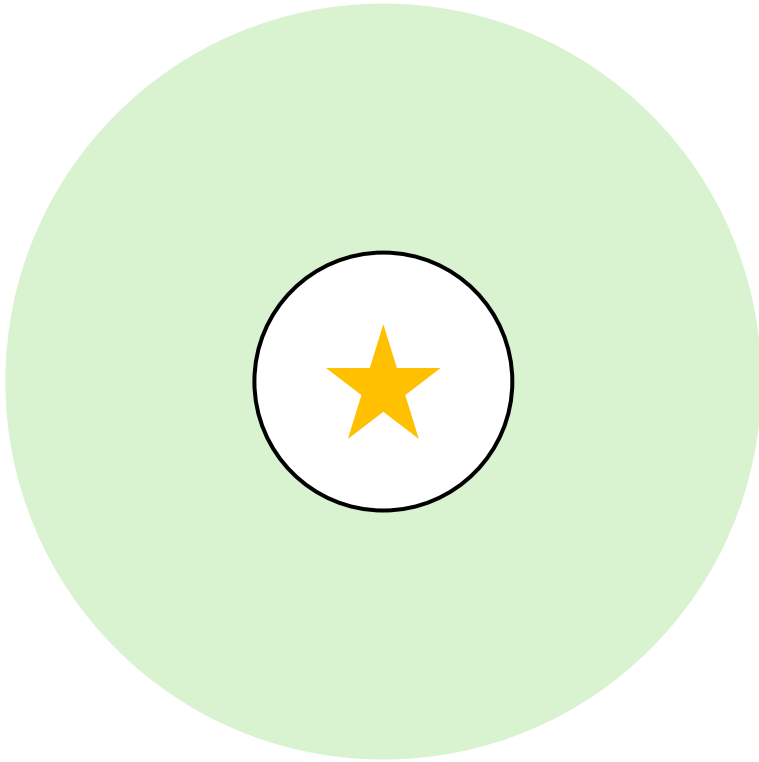
But not for all cavity modes



This triggers the overdamped modes...

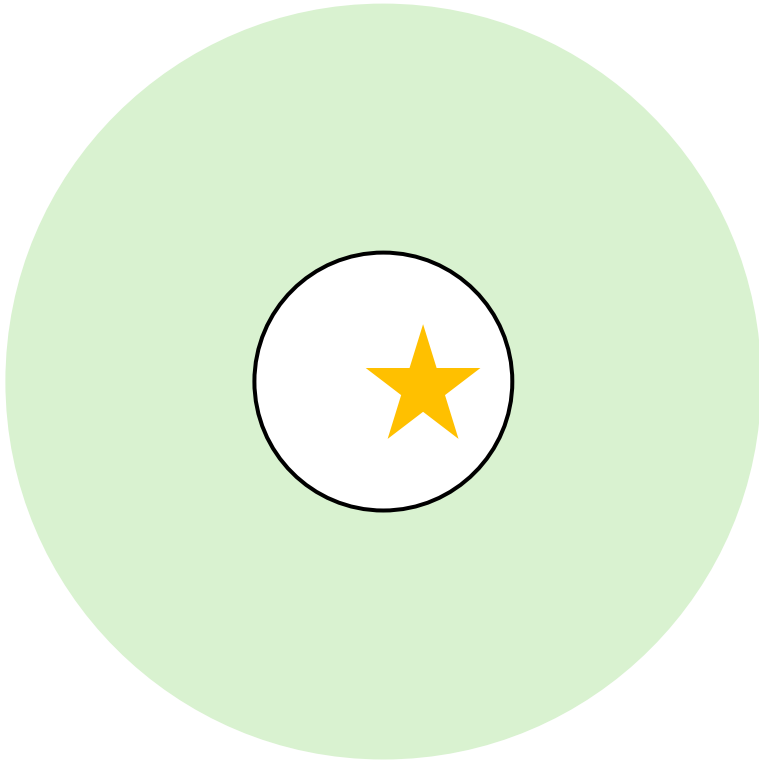
Catastrophic news: reflex modes are affected by the choice of reference frame

Barycentric frame



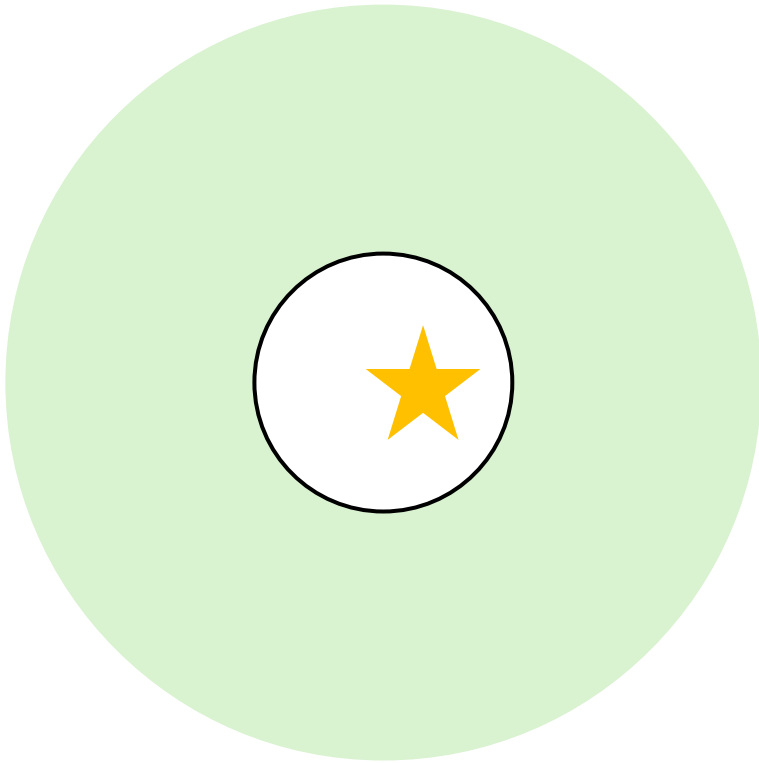
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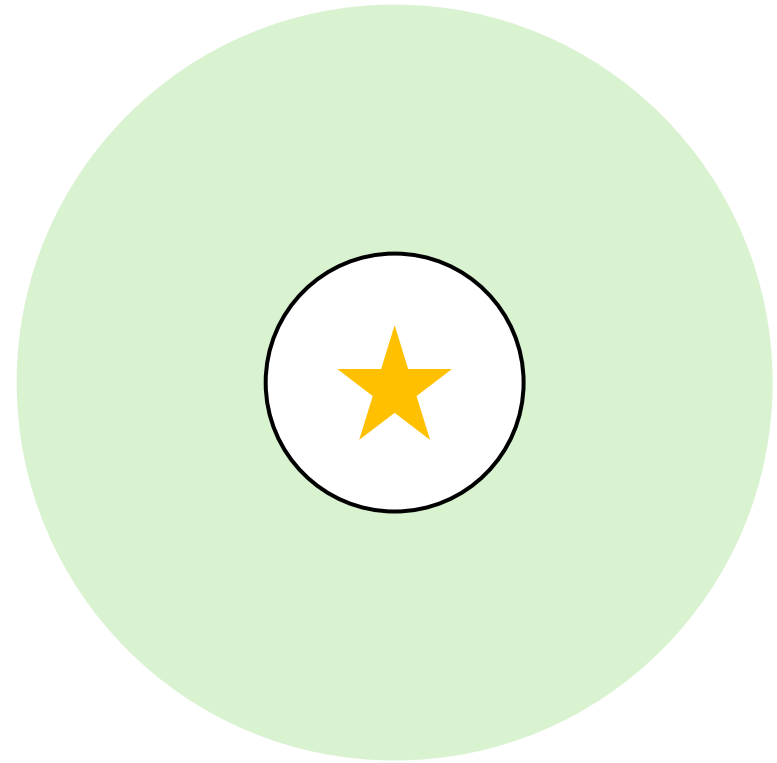


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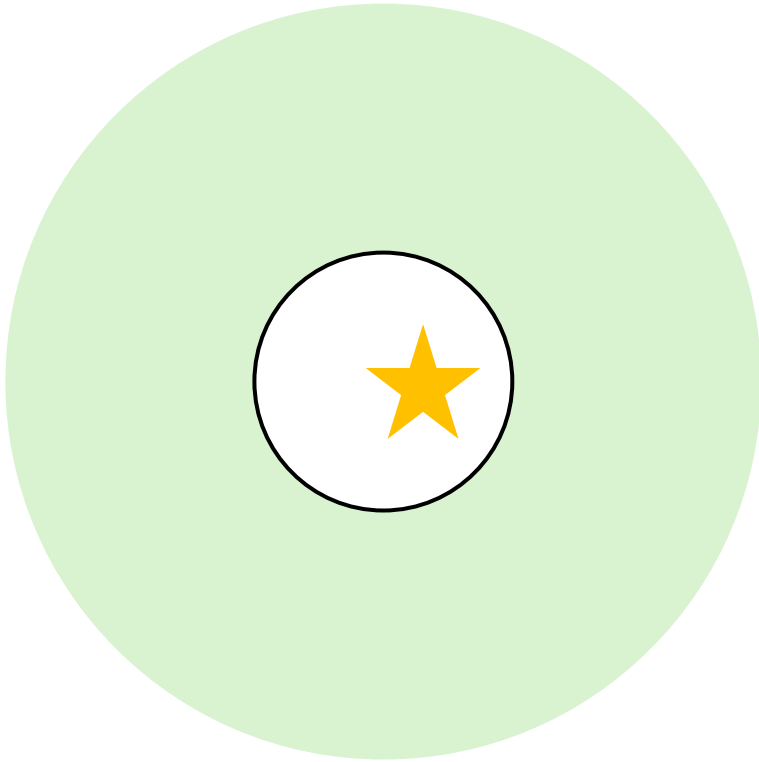


Stellocentric frame

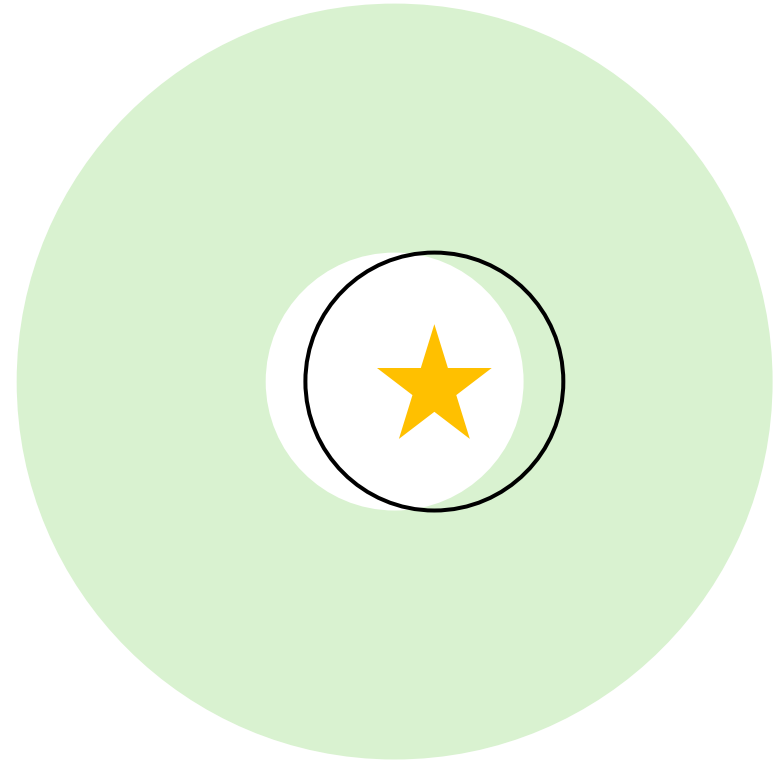


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Barycentric frame

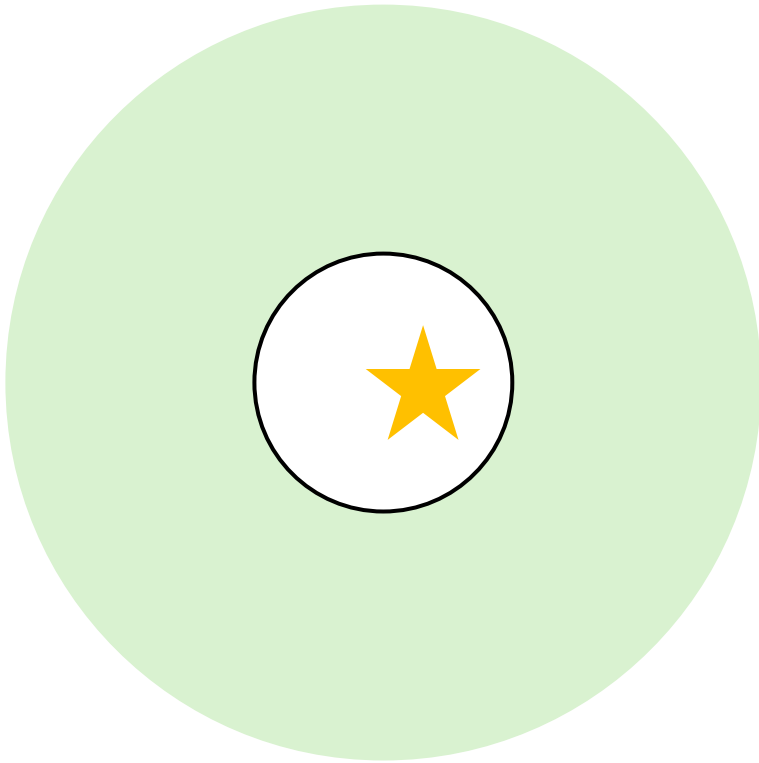


Stellocentric frame

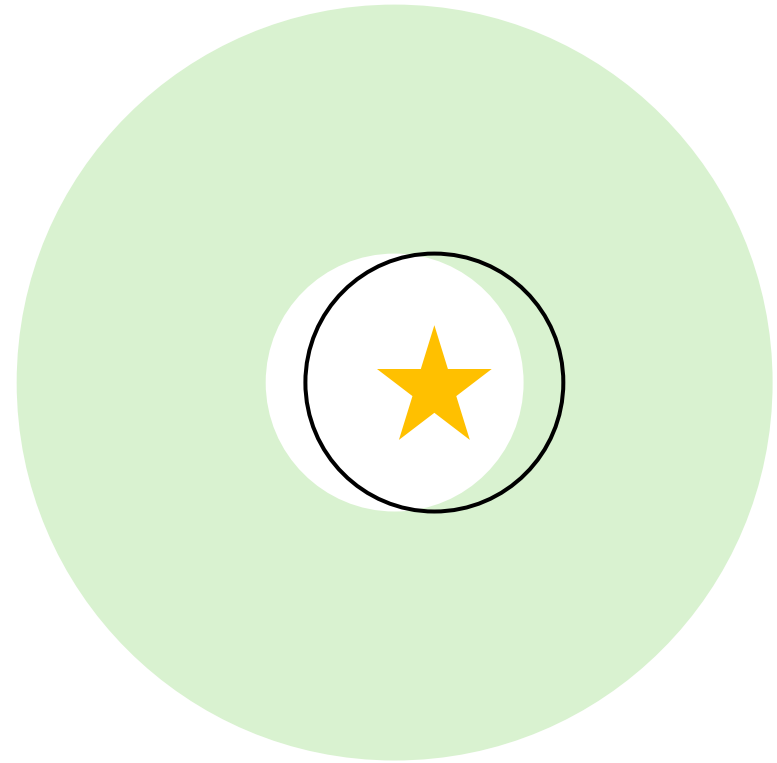


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Barycentric frame



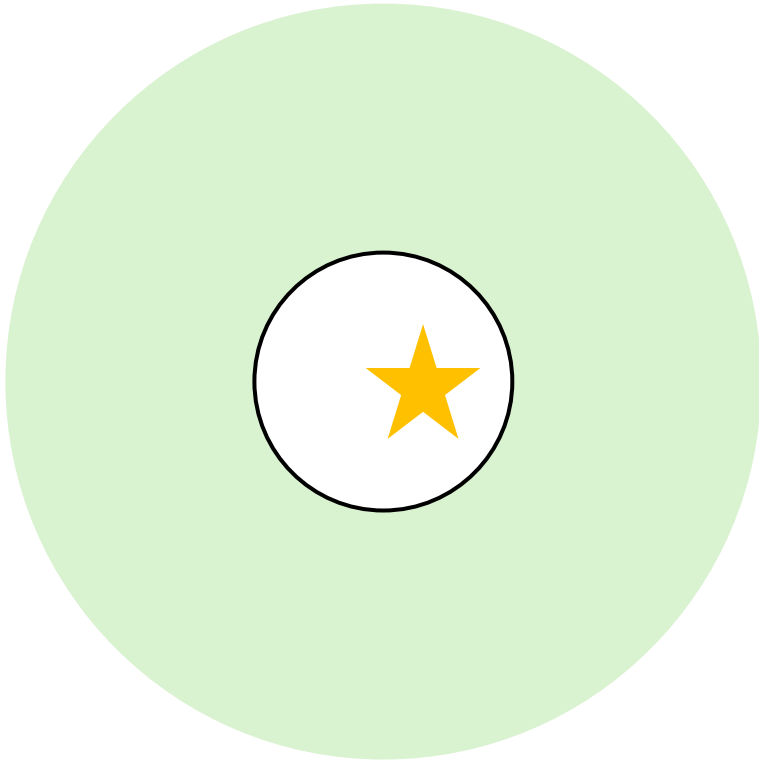
Stellocentric frame



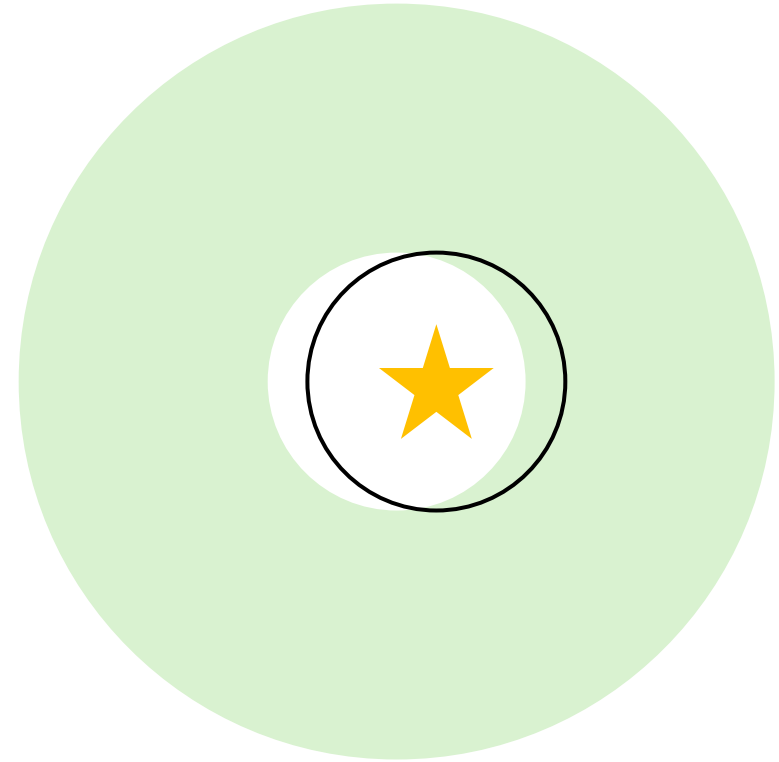
Boundaries (and damping layers) are not frame-invariant!

Catastrophic news: reflex modes are affected by the choice of reference frame

Barycentric frame



Stellocentric frame



Boundaries (and damping layers) are not frame-invariant!

The same issue affects the choice of EoM for the central object!

Take-home messages

- The reflex motion of the star can induce many waves and instabilities

This work was done in collaboration with H. Méheut, C. Baruteau, M. Tagger, J.-F. Gonzalez and A. Crida

Take-home messages

- The reflex motion of the star can induce many waves and instabilities
- As far as we can tell, they are all artefacts

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 - But that's prohibitively expensive
 - And there may not be an inner cavity nor a sharp outer edge...
 - Maybe we're not quite ready to model the reflex motion of the star

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 - Maybe we're not quite ready to model the reflex motion of the star
- Similar issues may affect other long-term, long-range perturbations
 - Fly-bys
 - External photoevaporation
 - Companions
 - Planets

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